Francesco Beretta
(CNRS / University of Bern / University of Neuchâtel)

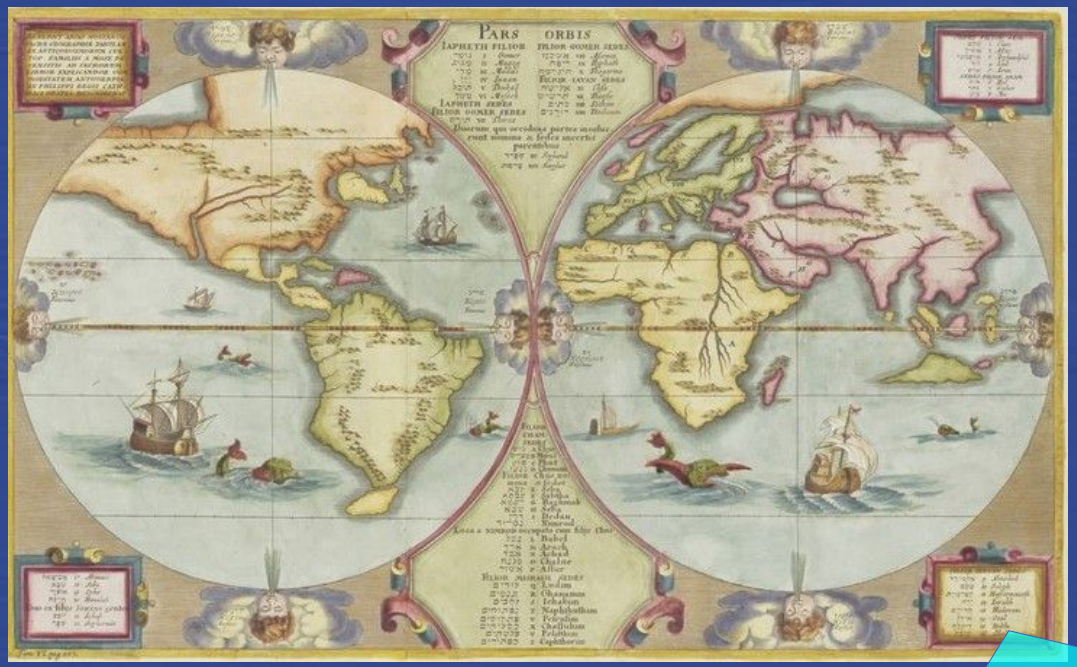
The semantic web and ontologies as a means of producing well-defined and reusable research data in the HSS

University of Bern
(version 1.0, october 2026)

Francesco Beretta, 13 October 2026 CC BY-NC-SA 4.0

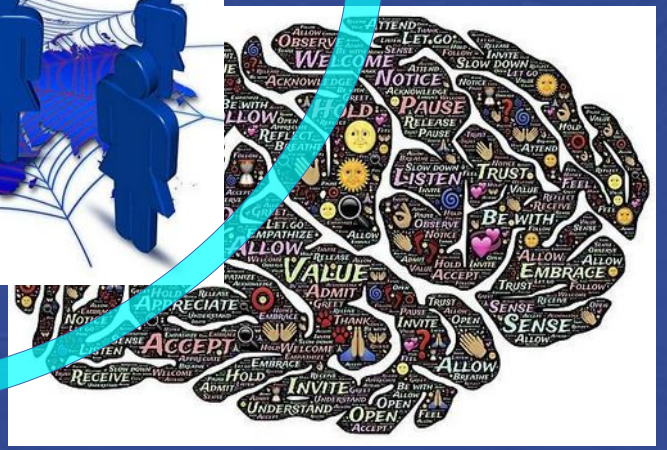
What are the key elements for building and managing information systems
in the humanities and social sciences (HSS) ?

Scientific knowledge as a socially defined, complex model of a domain of interest

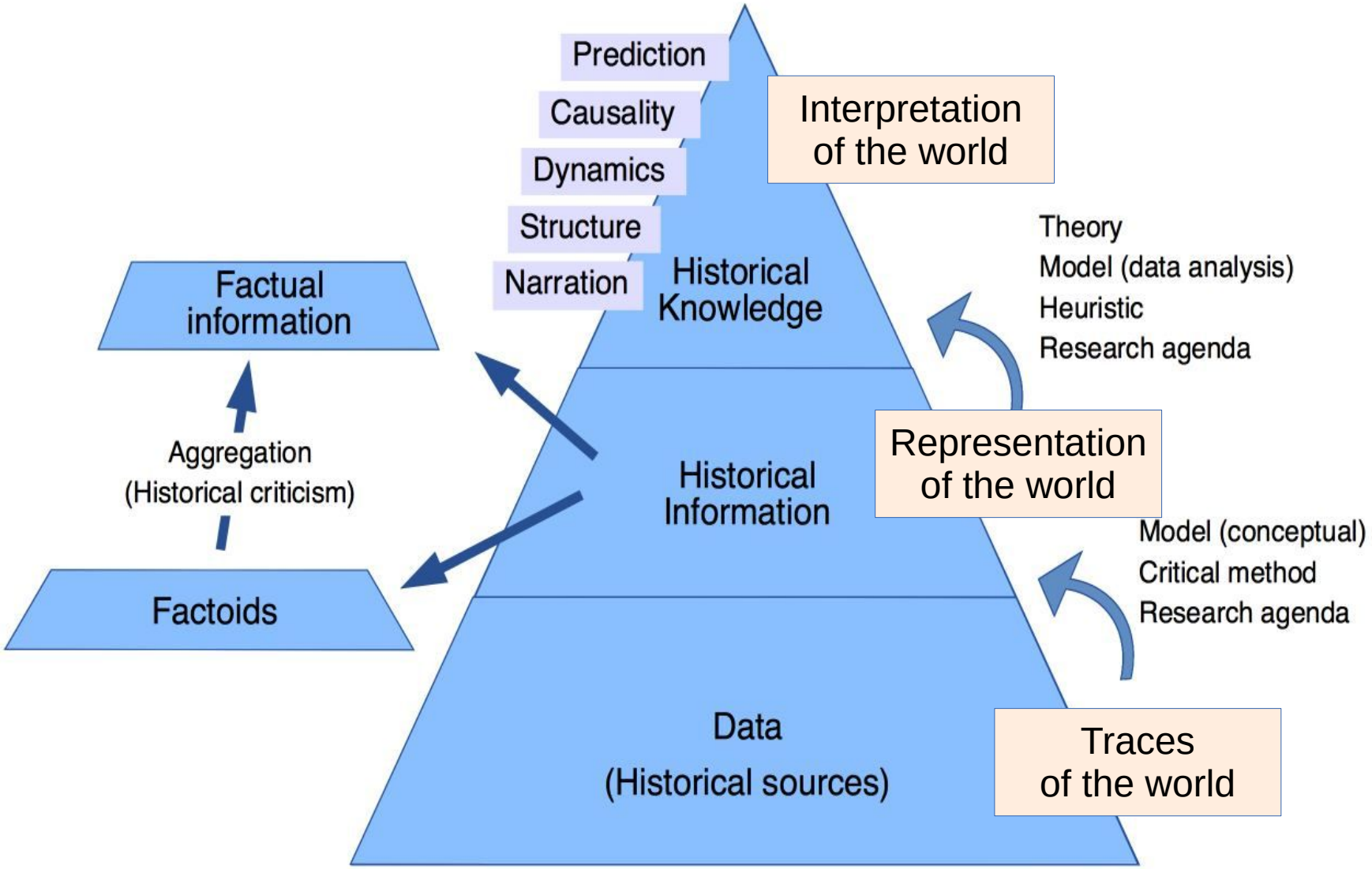








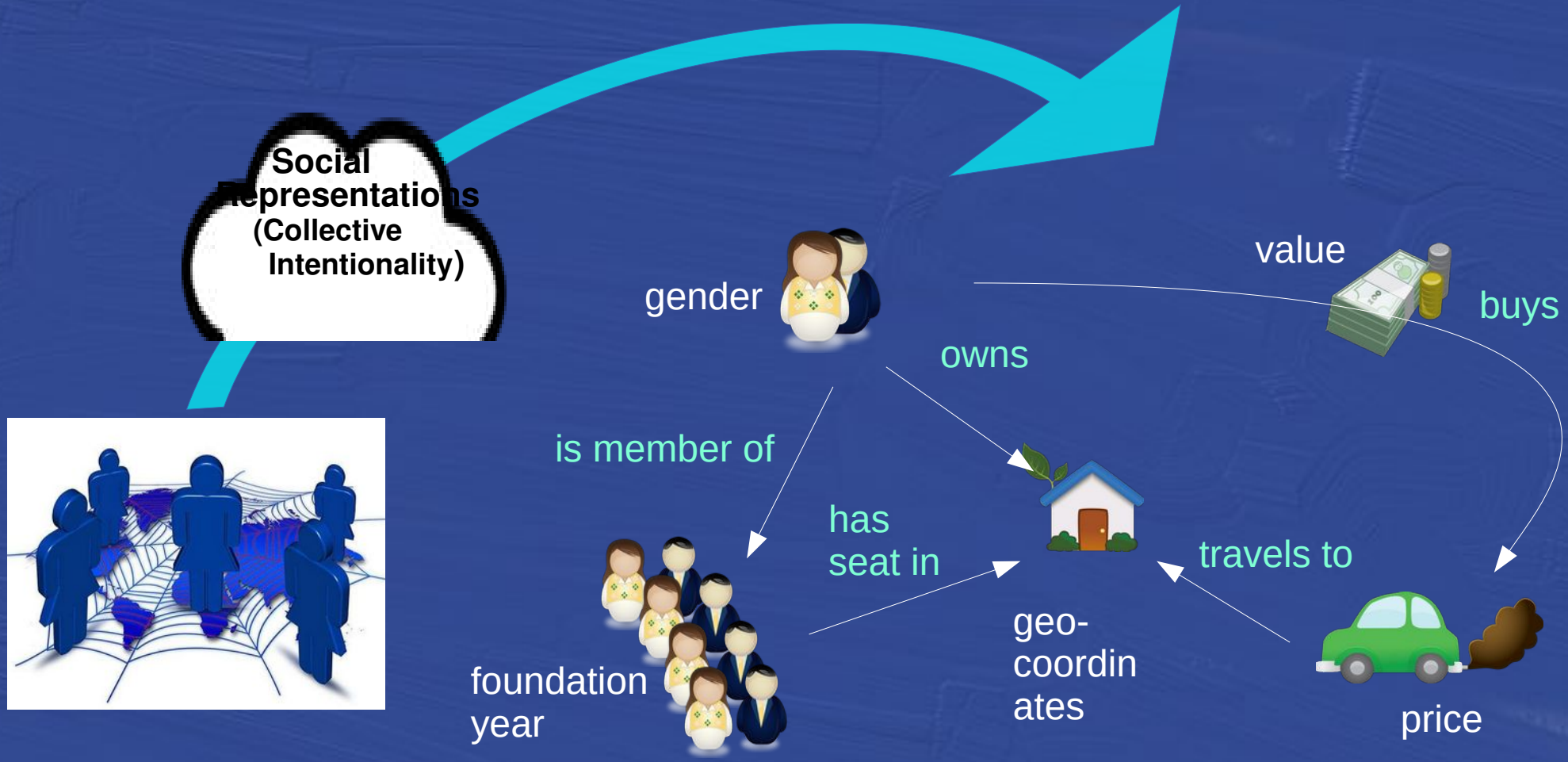
CC 0



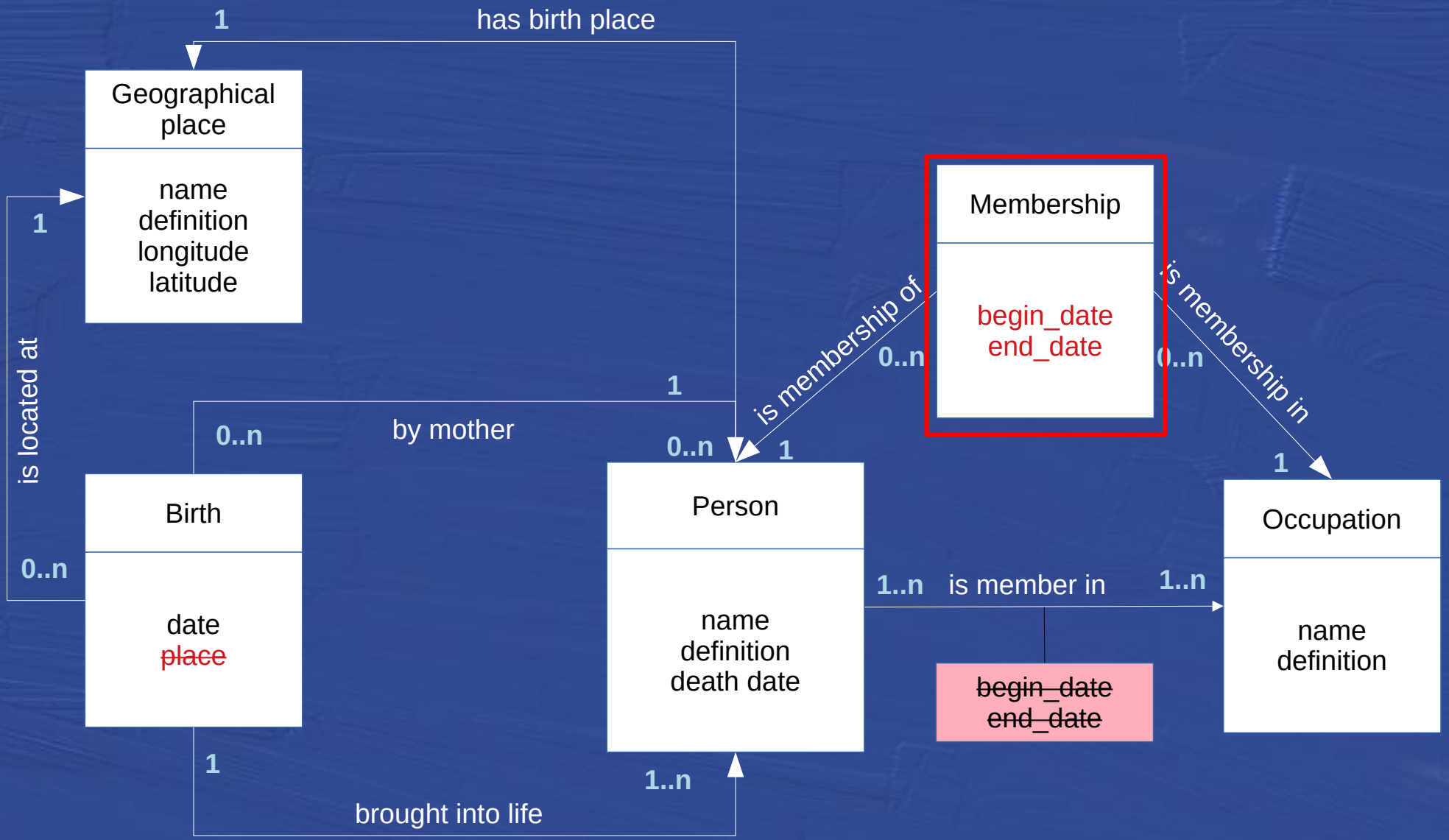
Francesco Beretta (CNRS/Université de Lyon), 7 July 2020 CC BY-NC-SA 4.0

Information as **representation** of the world :

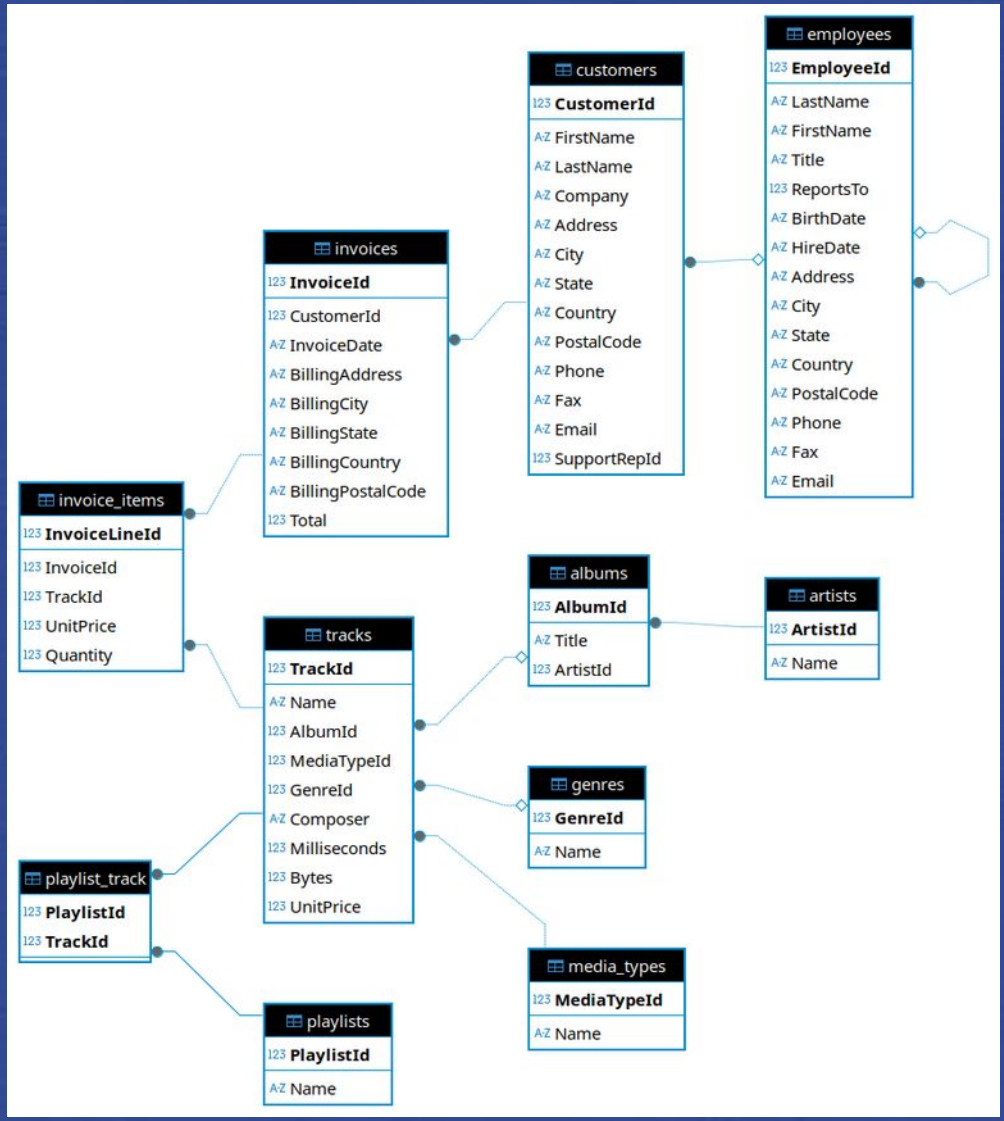
- representation of the **objects** in a domain of discourse
- of their **properties** (qualities)
- of their **relationships**



Conceptualizing the own research domain (domain of scientific discourse)



Relational databases as well established, robust technology for building information systems



Linked Open Data and the Semantic Web



Tim Berners-Lee, the inventor of the Web and Linked Data initiator, suggested a 5-star deployment scheme for Open Data.

https://5stardata.info/en/



★ make your stuff available **on the Web** (whatever format) under an open licence

★★ make it available as **structured data** (e.g., Excel instead of image scan of a table)

★★★ make it available in a non-proprietary **open format** (e.g., CSV instead of Excel)

★★★★ use **URIs to denote things**, so that people can point at your stuff

★★★★★ **link your data** to other data to provide context

Quoted from : https://5stardata.info/en/

Linked Open Data and the Semantic Web

(https://www.w3.org/TR/rdf11-concepts/)

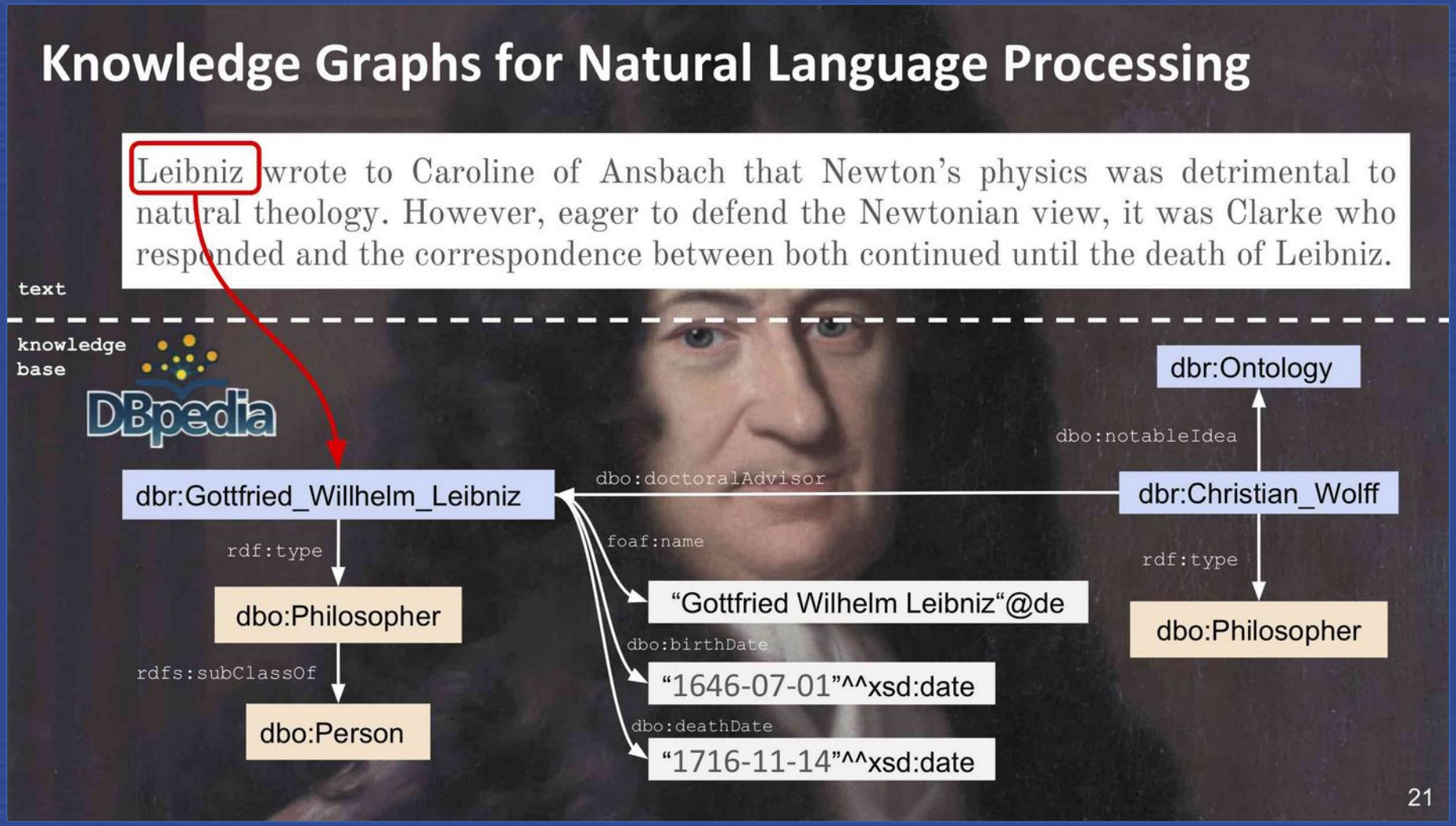
- « The Resource Description Framework (**RDF**) is a framework for **representing information in the Web**. »

- « A graph-based data model »



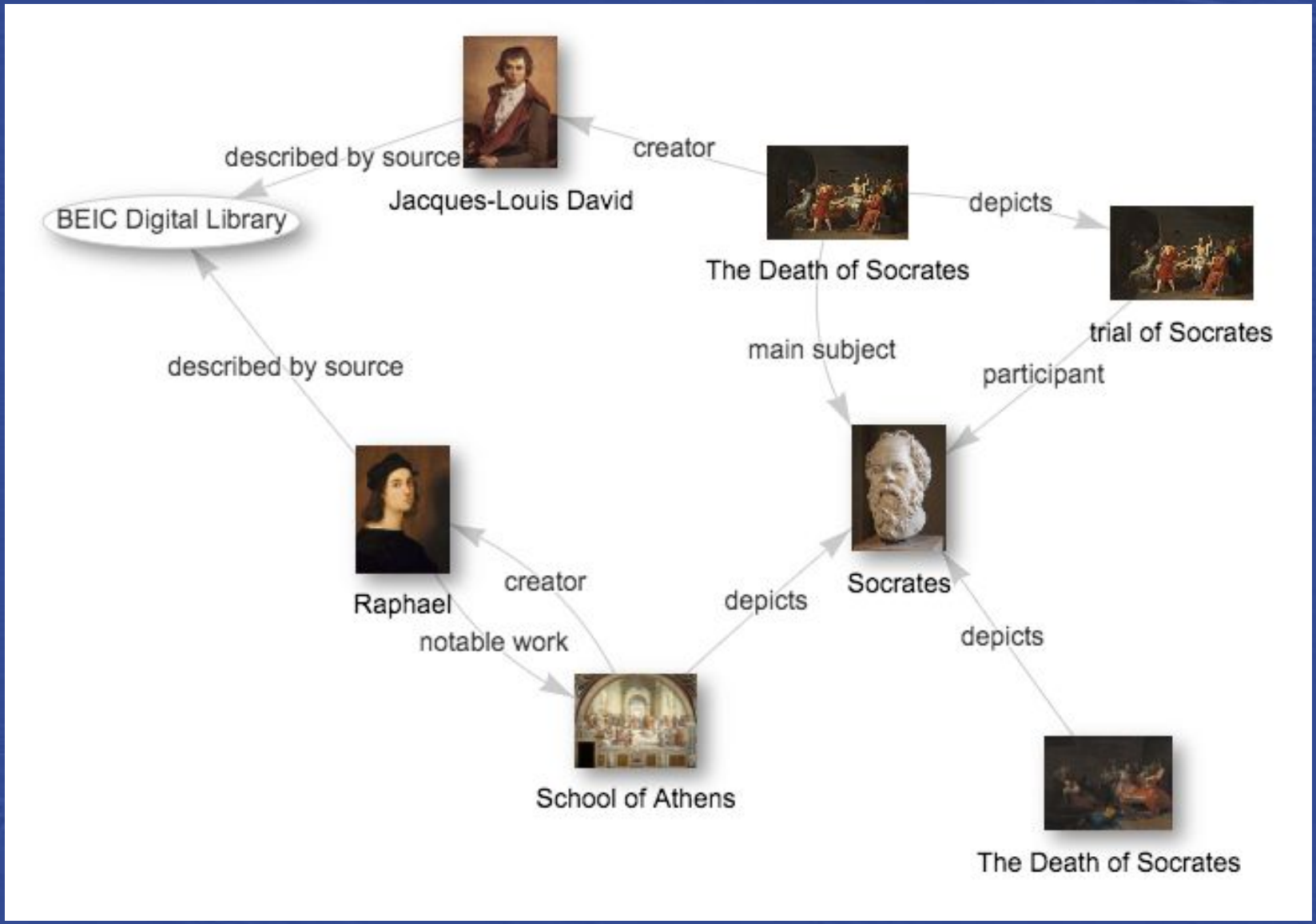
RDF triple

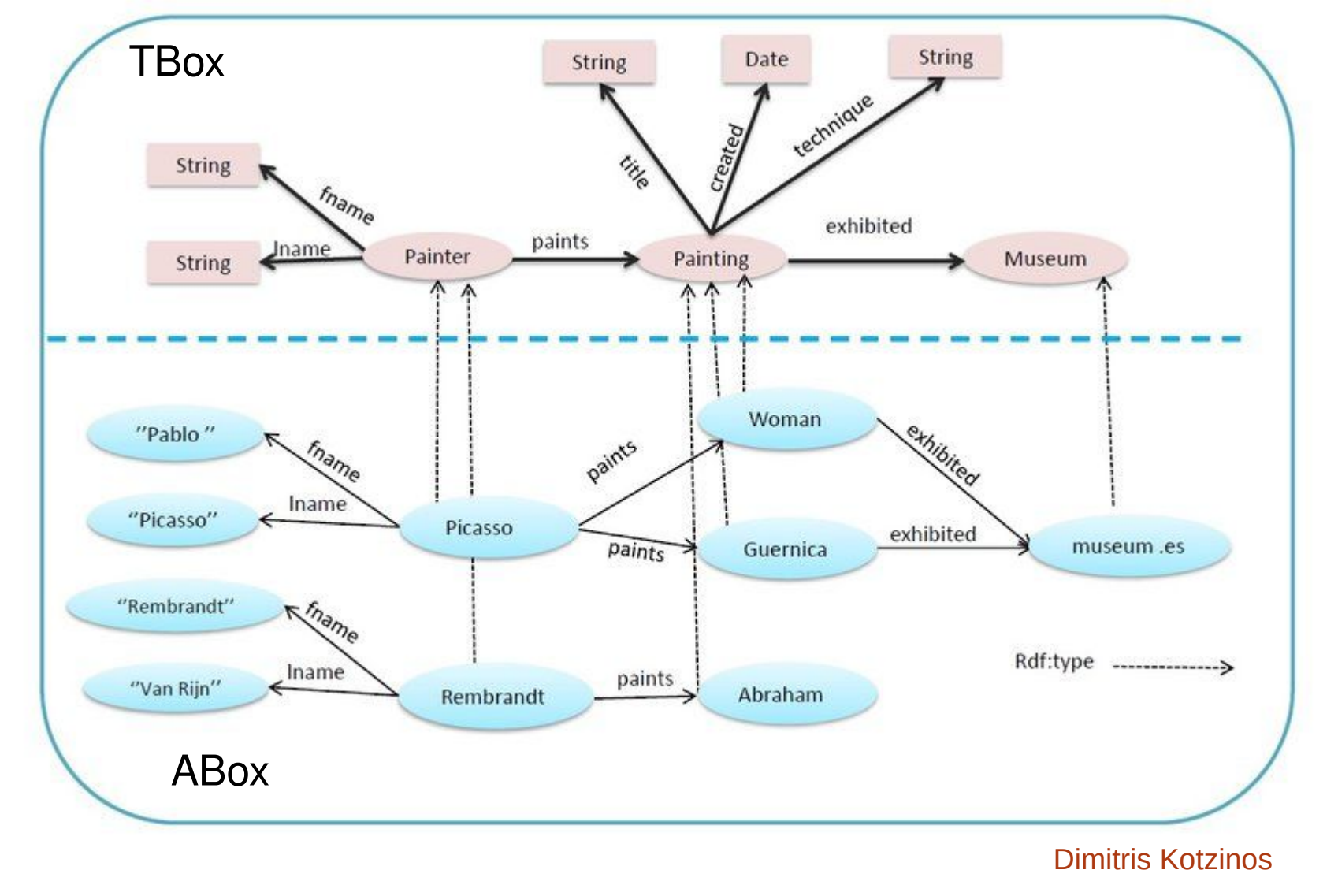
Where are the tables? In the graph !



Prof. Dr. Harald Sack, Antrittsvorlesung
Karlsruher Institut für Technologie (29.11.2017)

Wikidata 👈 An *information* graph (‘knowledge graph’) representing the objects in the world and their relationships





Dimitris Kotzinos

- ABox (Assertional Box): information graph, instance-level data
- TBox (Terminological Box): schema-level rules, RDFS / OWL-DL / SCHACL

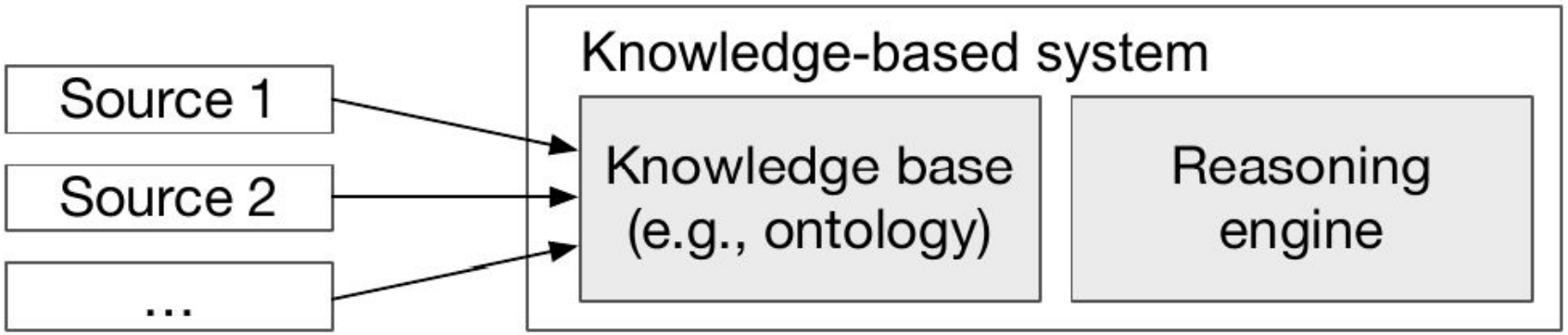
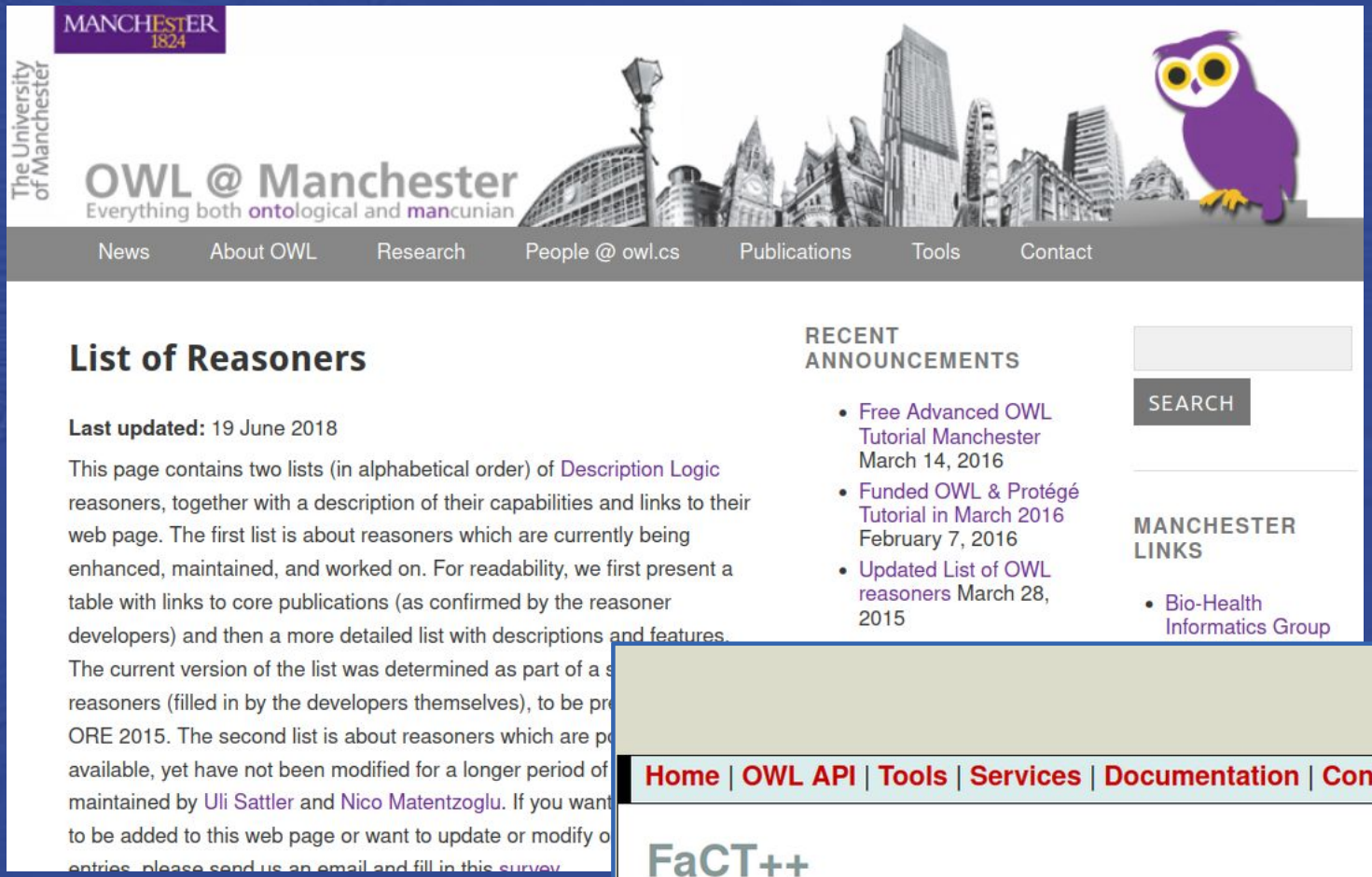
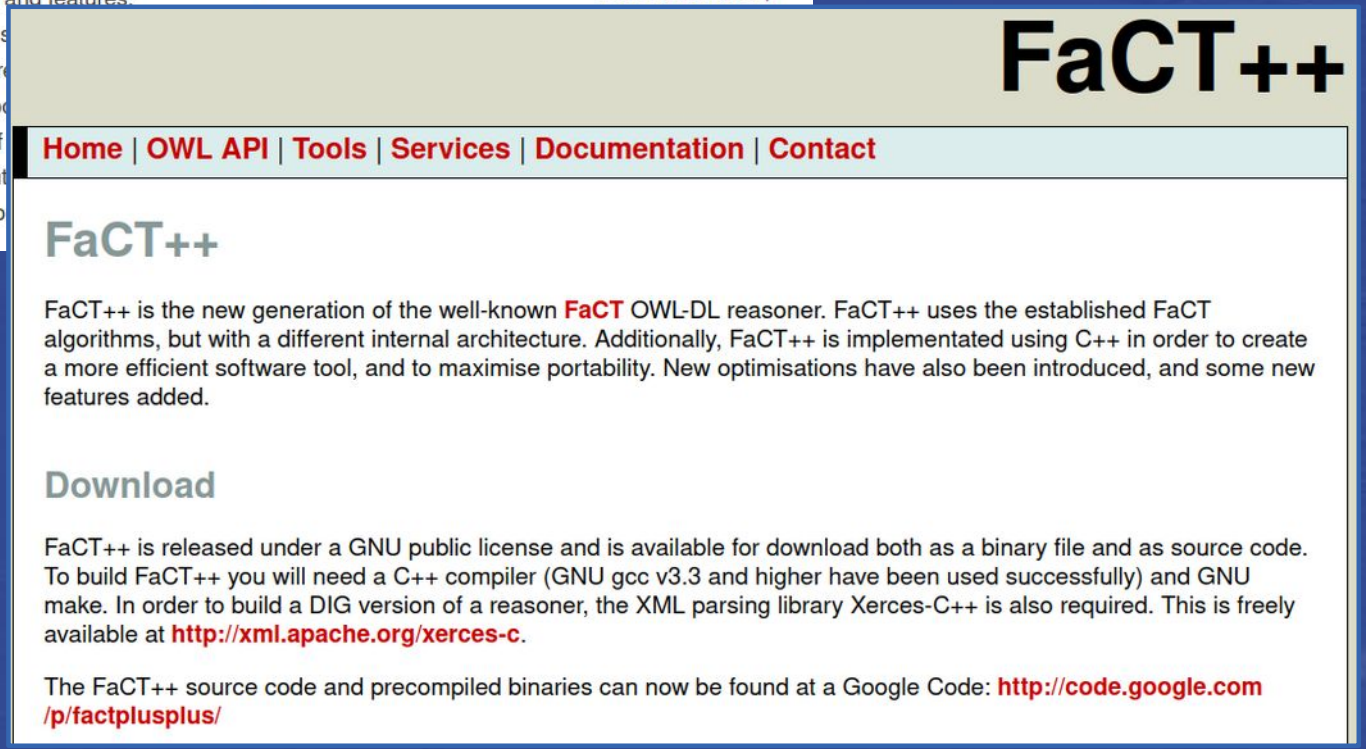


Figure 1: Architecture of a knowledge graph

Ehrlinger, Lisa and Wolfram Wöß, 'Towards a Definition of Knowledge Graphs',

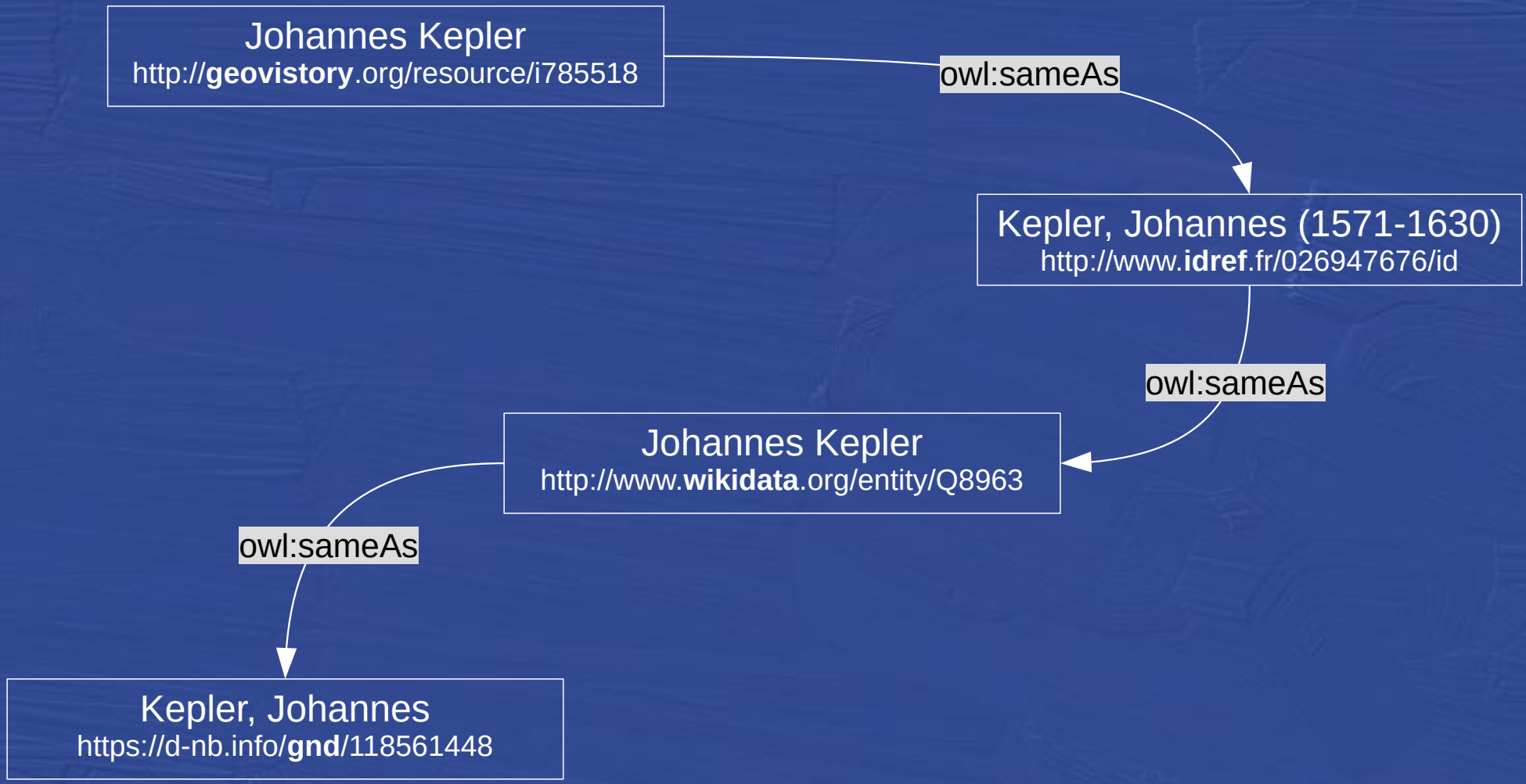
Joint Proceedings of the Posters and Demos Track of the 12th International Conference on Semantic Systems ..., Leipzig, Germany, September 12-15, 2016, 2016 (https://ceur-ws.org/Vol-1695/paper4.pdf)


MANCHESTER 1824
The University of Manchester
OWL @ Manchester
Everything both ontological and mancunian
News
About OWL
Research
People @ owl.cs
Publications
Tools
Contact
List of Reasoners
Last updated: 19 June 2018
This page contains two lists (in alphabetical order) of Description Logic reasoners, together with a description of their capabilities and links to their web page. The first list is about reasoners which are currently being enhanced, maintained, and worked on. For readability, we first present a table with links to core publications (as confirmed by the reasoner developers) and then a more detailed list with descriptions and features.
The current version of the list was determined as part of a s
reasoners (filled in by the developers themselves), to be pr
ORE 2015. The second list is about reasoners which are p
available, yet have not been modified for a longer period of
maintained by Uli Sattler and Nico Matentzoglu. If you want
to be added to this web page or want to update or modify o
entries, please send us an email and fill in this survey
RECENT ANNOUNCEMENTS
Free Advanced OWL Tutorial Manchester March 14, 2016
Funded OWL & Protégé Tutorial in March 2016 February 7, 2016
Updated List of OWL reasoners March 28, 2015
SEARCH
MANCHESTER LINKS
Bio-Health Informatics Group


FaCT++
Home | OWL API | Tools | Services | Documentation | Contact
FaCT++
FaCT++ is the new generation of the well-known FaCT OWL-DL reasoner. FaCT++ uses the established FaCT algorithms, but with a different internal architecture. Additionally, FaCT++ is implementated using C++ in order to create a more efficient software tool, and to maximise portability. New optimisations have also been introduced, and some new features added.
Download
FaCT++ is released under a GNU public license and is available for download both as a binary file and as source code. To build FaCT++ you will need a C++ compiler (GNU gcc v3.3 and higher have been used successfully) and GNU make. In order to build a DIG version of a reasoner, the XML parsing library Xerces-C++ is also required. This is freely available at http://xml.apache.org/xerces-c.
The FaCT++ source code and precompiled binaries can now be found at a Google Code: http://code.google.com /p/factplusplus/

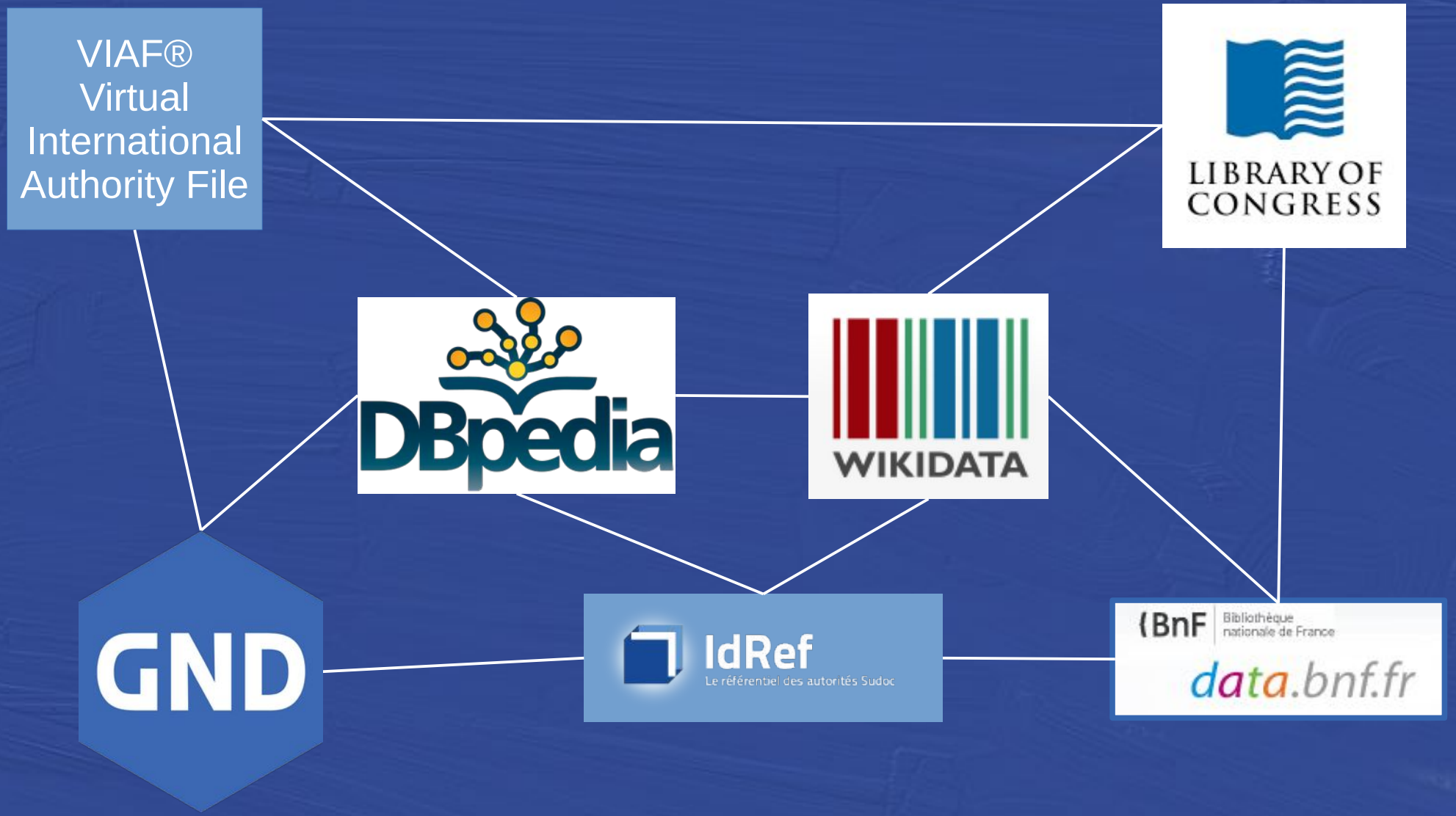
LOD as Links

« URIs to denote things, so that people can point at your stuff »

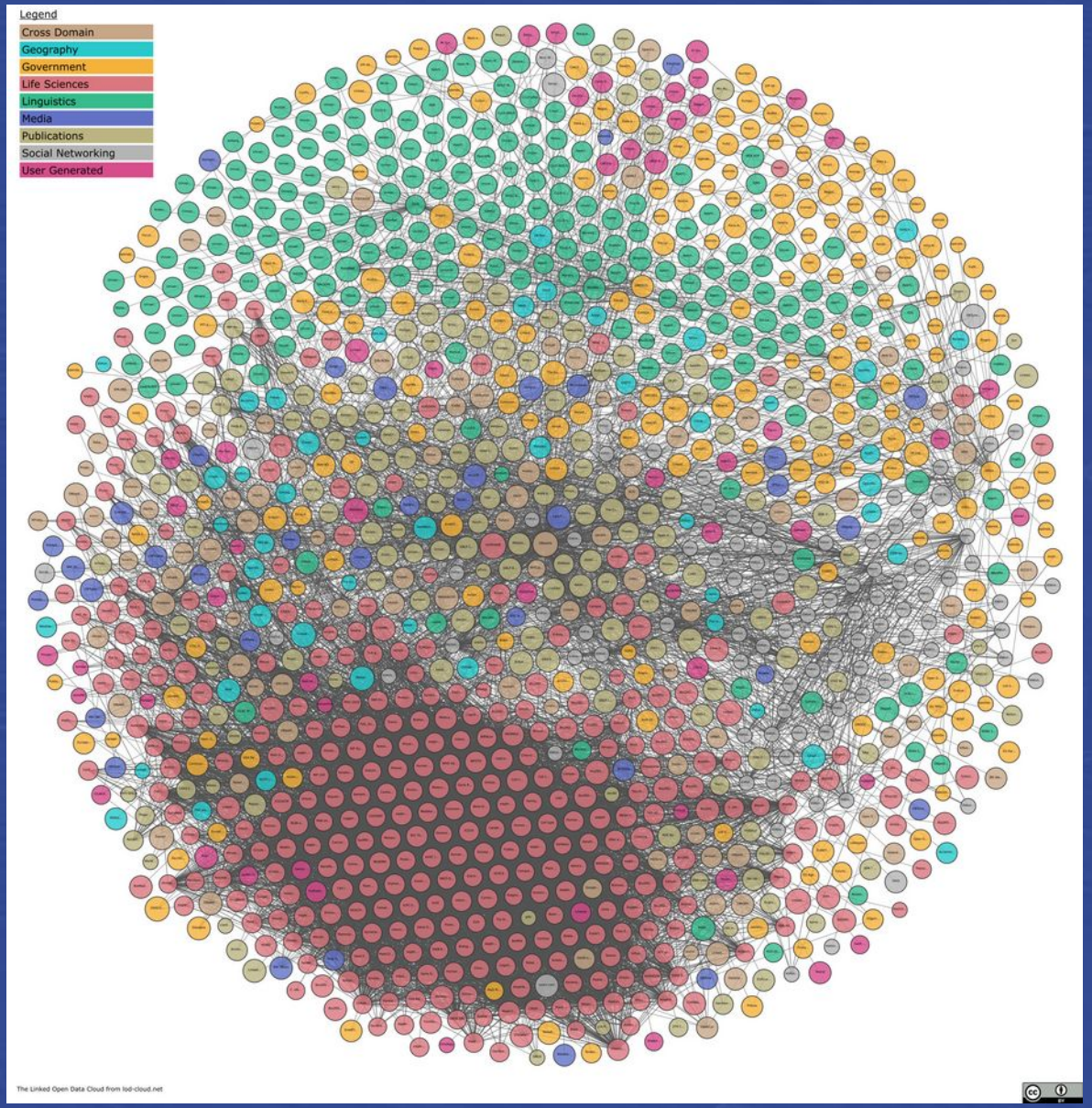


LOD as Links

« URIs to denote things, so that people can point at your stuff »

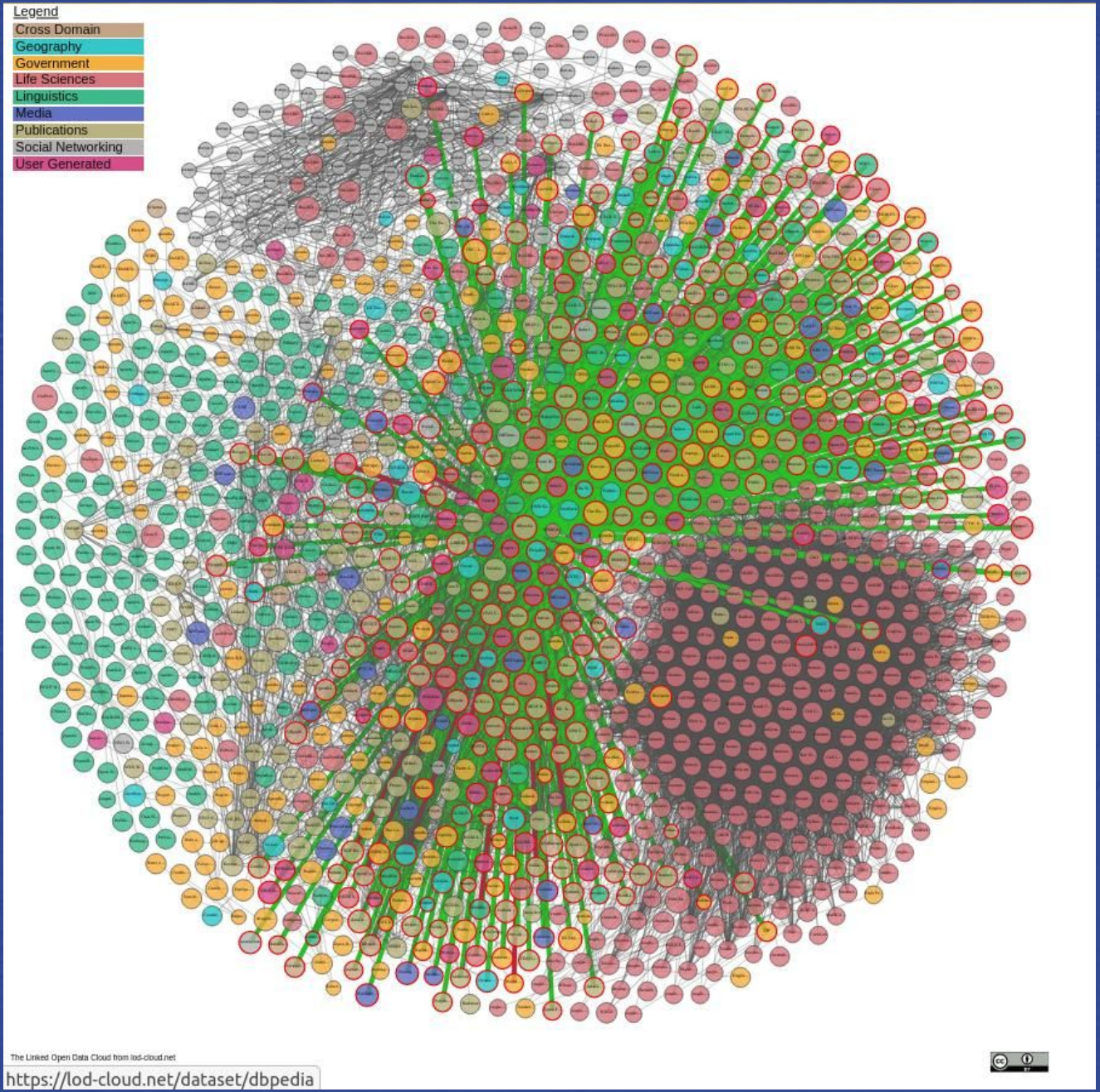


Semantic Web

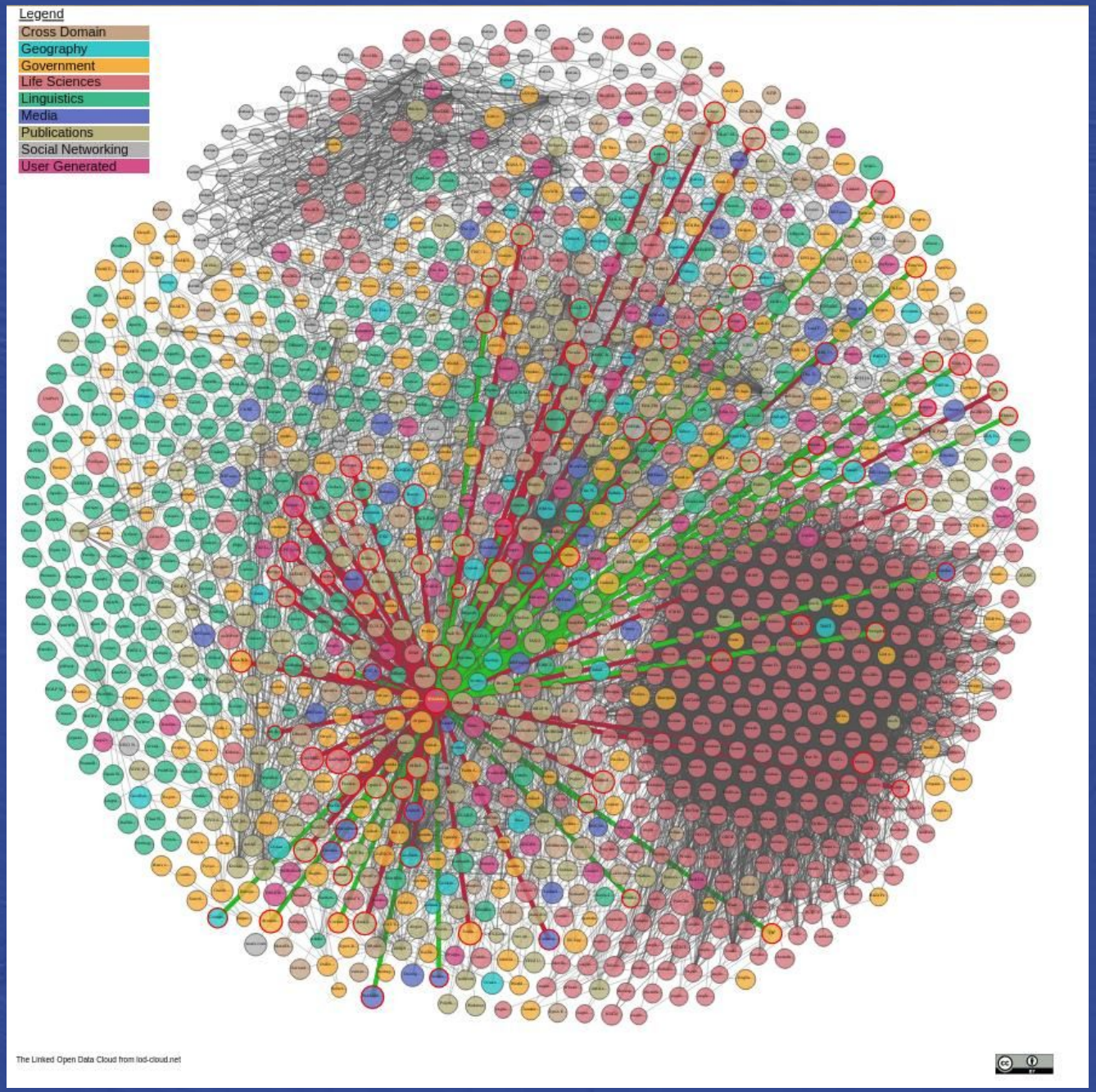


https://lod-cloud.net/

DBPedia


Legend
Cross Domain
Geography
Government
Life Sciences
Linguistics
Media
Publications
Social Networking
User Generated
The Linked Open Data Cloud from lod-cloud.net
https://lod-cloud.net/dataset/dbpedia

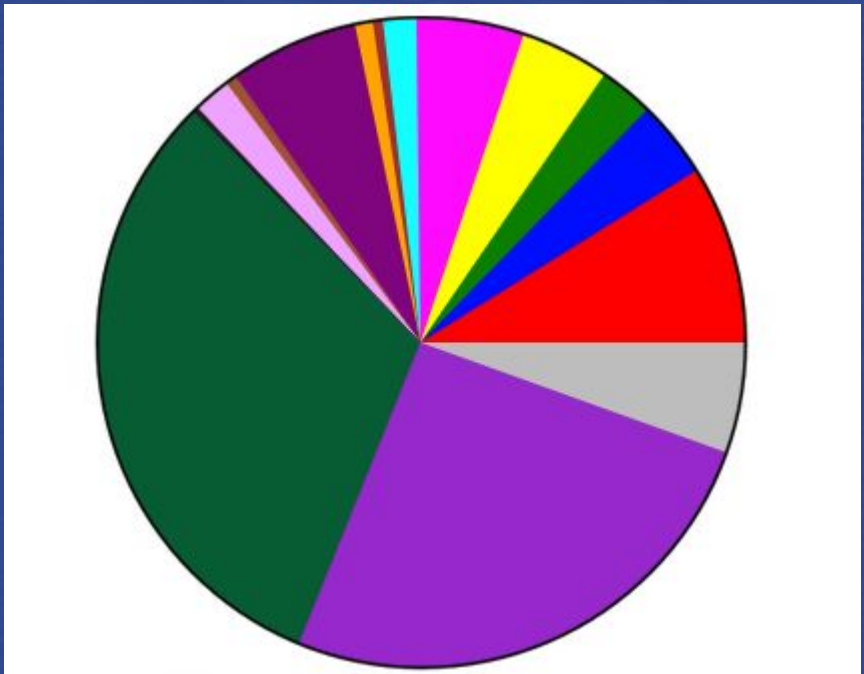
Wikidata


Legend
Cross Domain
Geography
Government
Life Sciences
Linguistics
Media
Publications
Social Networking
User Generated
The Linked Open Data Cloud from lod-cloud.net

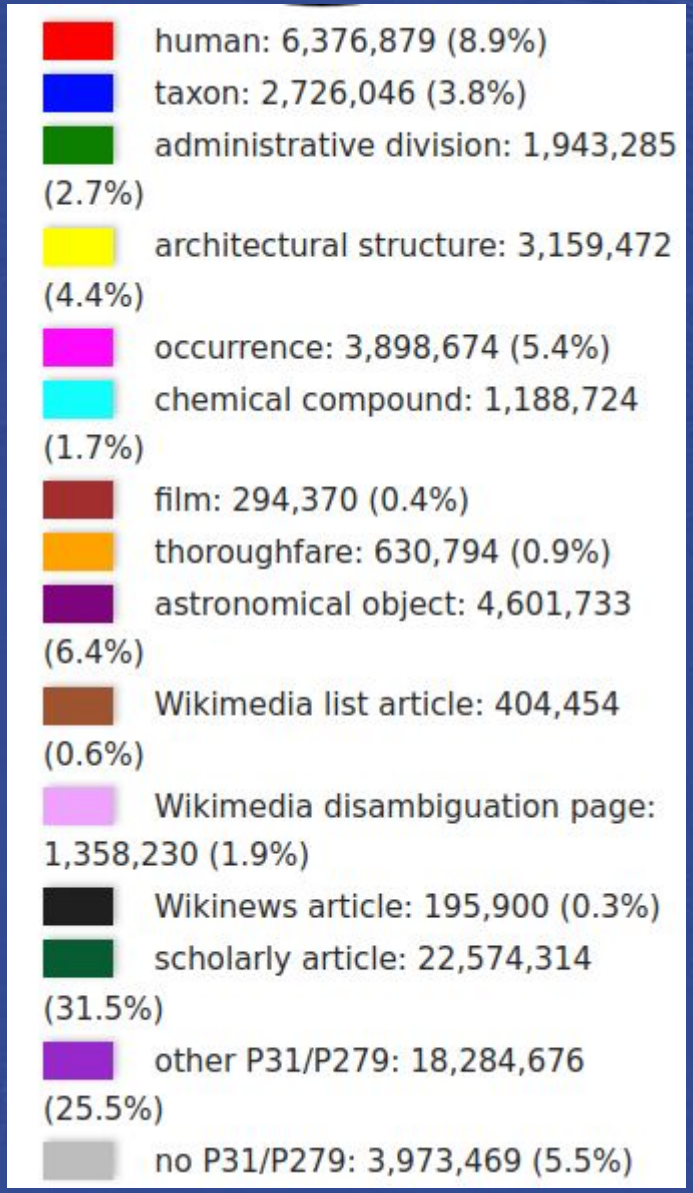
Wikidata

14 November 2023 – 107’588’216 items

1.5 billions statements

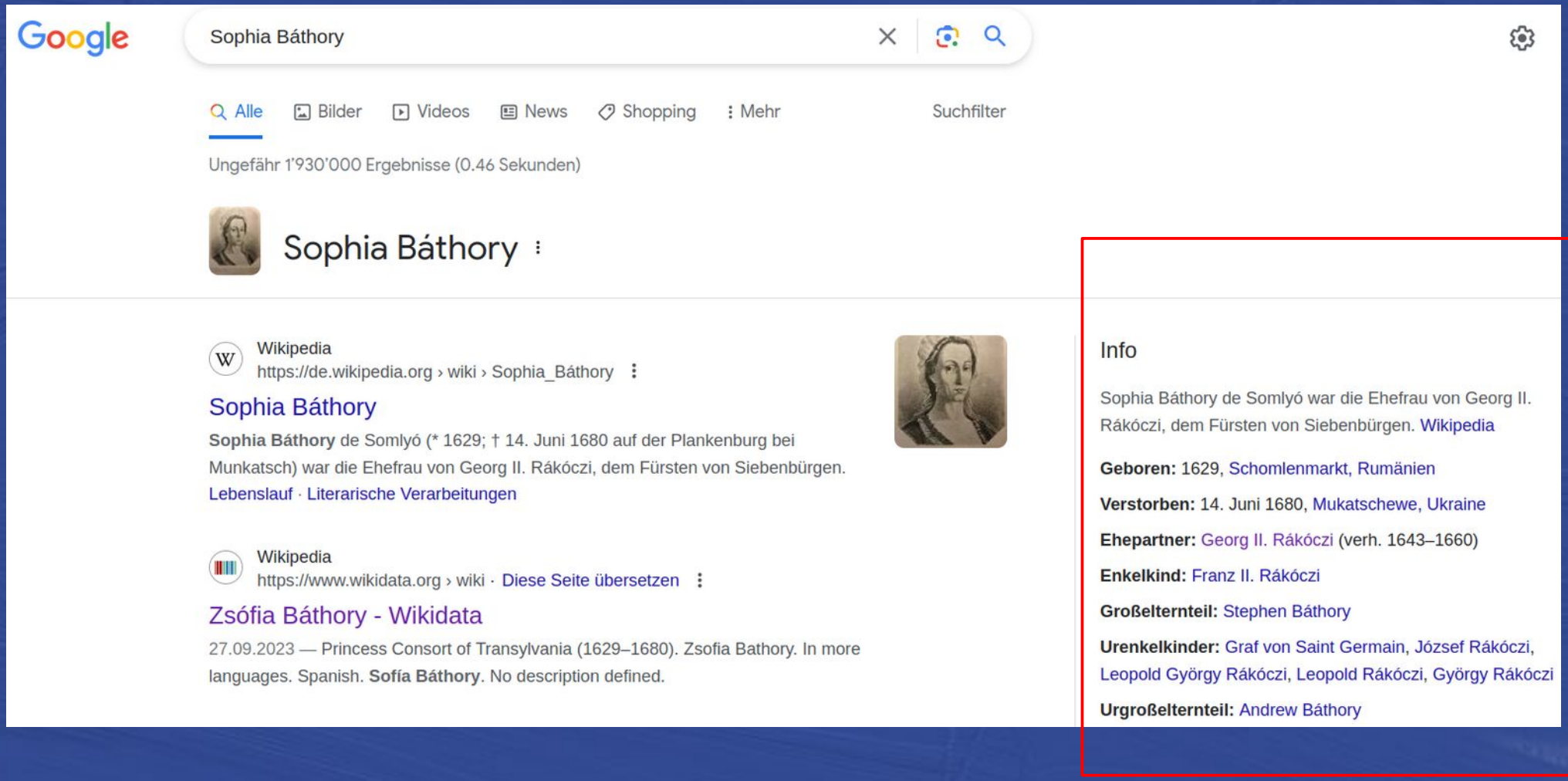


16 February 2020 : 71,611,020 items



Google Knowledge Graph

"By March 2023, it had grown to 800 billion facts on 8 billion entities" (Wikipedia).



Wikidata and Google Knowledge Graph

WIKIDATA

Item | Discussion

Zsófia Báthory (Q250942)

Princess Consort of Transylvania (1629–1680)
Zsofia Bathory

edit

In more languages
Configure

Language	Label	Description	Also known as
English	Zsófia Báthory	Princess Consort of Transylvania (1629–1680)	Zsofia Bathory
German	Zsófia Báthory	Ehefrau von Georg II. Rákóczi, des Fürsten von Siebenbürgen (1629–1680)	
Alemannic	No label defined	No description defined	
French	Zsófia Báthory	(1629–1680)	

All entered languages

Main page
Community portal
Project chat
Create a new Item
Recent changes
Random Item
Query Service
Nearby
Help
Donate

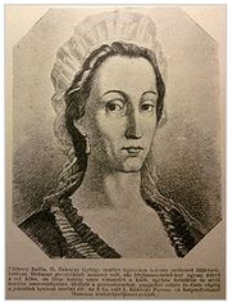
Lexicographical data
Create a new Lexeme
Recent changes
Random Lexeme

Tools
What links here
Related changes
Special pages
Permanent link
Page information
Concept URI
Cite this page
Get shortened URL

Statements

instance of — human
1 reference

image



Báthory Zsófia 1629.jpg
585 × 779; 155 KB

Google Knowledge Graph ID — /g/121258kx — edit
0 references
+ add reference
+ add value

Hungarian National Namespace person ID (new) — 662639 — edit
0 references
+ add reference
+ add value

WIKIDATA

Property | Discussion

Google Knowledge Graph ID (P2671)

identifier for Google Knowledge Graph API, starting with "/g/". For IDs starting with "/m/", use Freebase ID (P646)

In more languages
Configure

Language	Label	Description	Also known as
English	Google Knowledge Graph ID	identifier for Google Knowledge Graph API, starting with "/g/". For IDs starting with "/m/", use Freebase ID (P646)	

Main page
Community portal
Project chat
Create a new Item
Recent changes
Random Item
Query Service
Nearby

From a resource-centered model...



...to a statement-centered model...



WIKIDATA

BNF Authorities

statement

"1571-12-21"

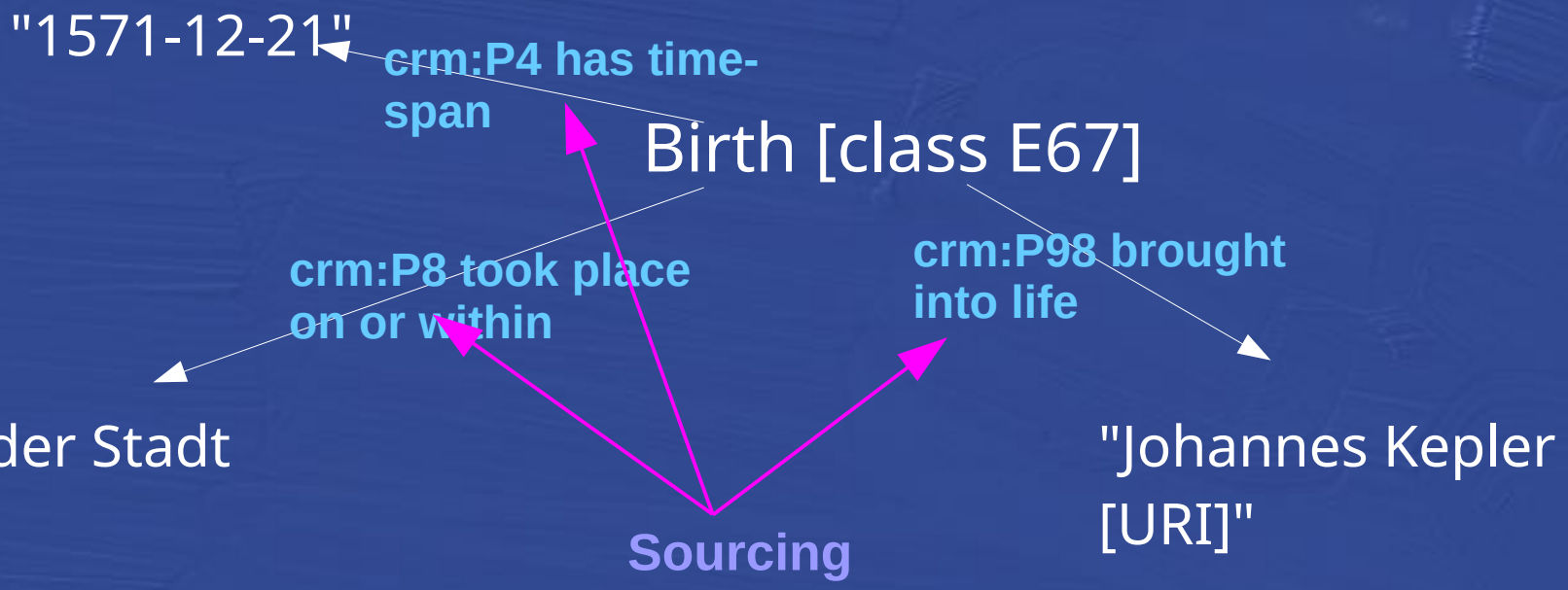
wdt:P569 (date of birth)

Kepler, Johannes (Q8963)

wdt:P19 (place of birth)

Weil der Stadt (Q505147)

… or prefer a more robust event-centered model !



Interoperability

Findable

Accessible

Interoperable

Re-usable

«There is an urgent need to improve the infrastructure supporting the ***reuse*** of scholarly data »

Wilkinson, Mark D., Michel Dumontier, Ijsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, et al. "*The FAIR Guiding Principles for Scientific Data Management and Stewardship*." Scientific Data 3 (March 15, 2016): 160018.

The FAIR Data Principles

To be **Interoperable**:

I1. (meta)data use a *formal, accessible, shared, and broadly applicable* ***language*** *for* ***knowledge representation***.

I2. (meta)data use *vocabularies that follow FAIR principles*.

I3. (meta)data include qualified references to other (meta)data.

To be **Re-usable**:

R1. meta(data) have a plurality of accurate and relevant attributes.

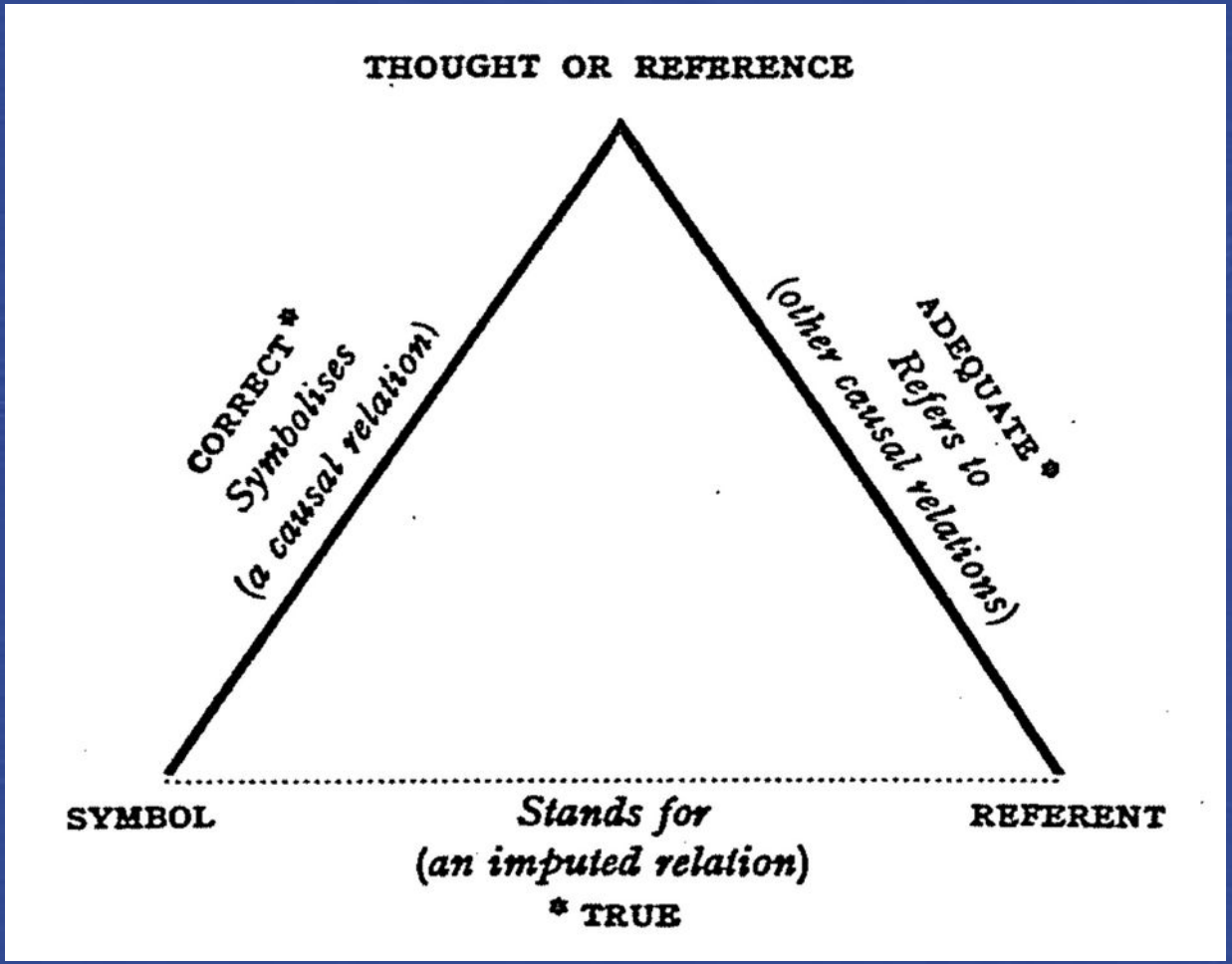
R1.1. (meta)data are released with a *clear and accessible data usage license*.

R1.2. (meta)data are associated with their *provenance*.

R1.3. (meta)data meet ***domain-relevant community standards***.

Ontology - ontologies

Ontologies are about meaning: the semantic web



Triangle of reference :

- URIs denote things
- Concepts (classes) connote things

“An ontology is
a formal explicit specification
of a shared conceptualization
of a domain of interest”

- « Formality – ... a knowledge representation language that is based on the grounds of **formal semantics**. »
- « Consensus – … an agreement on a domain conceptualization among people in a community. »
- « Conceptuality – … in terms of conceptual symbols that can be intuitively grasped by humans, as they correspond to the elements in their **mental models**. »
- « Domain Specificity – ... limited to knowledge about a particular **domain of interest**. »

[Domingue et al. 2011, p. 510-511]

DOLCE Lite + DUL

CIDOC CRM

SDHSS

schema.org

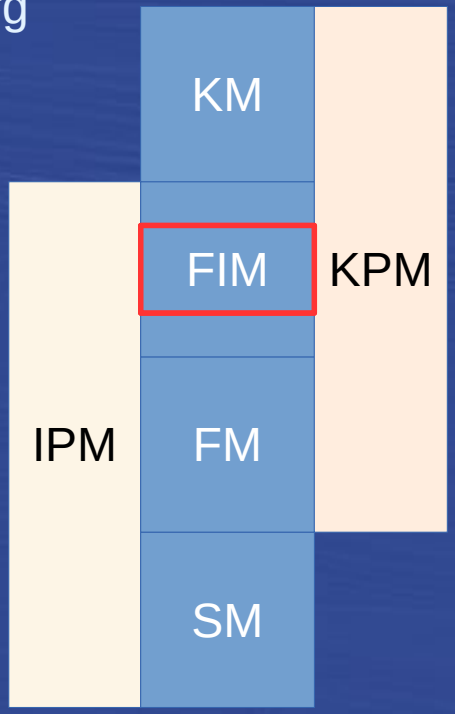
DBpedia Ontology

Wikidata Ontology

HiCO

PROV

Web Annotation Data Model



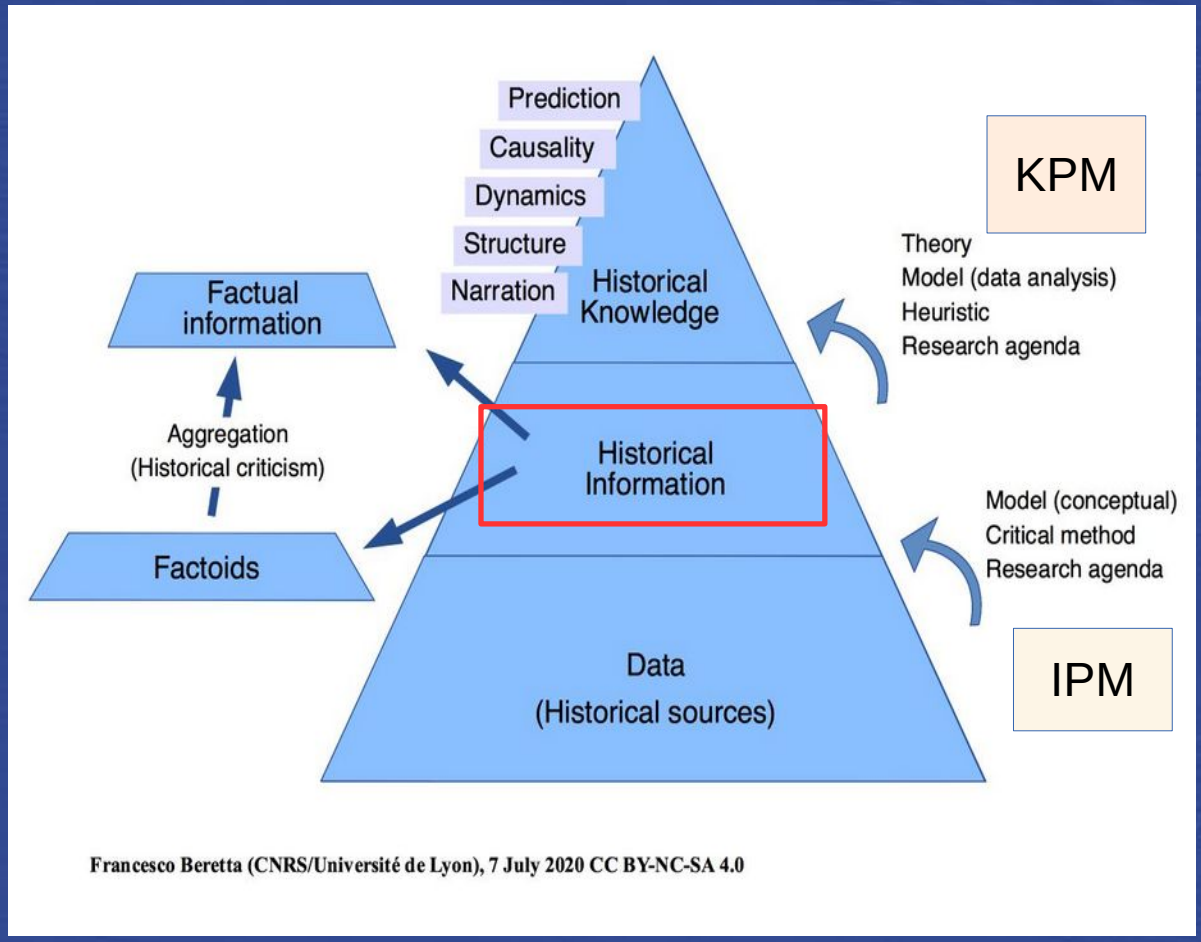
FRBR LRM

EAD

TEI

RiC

DCMI



CIDOC CRM

DUL
(DOLCE ULTRA LIGHT)

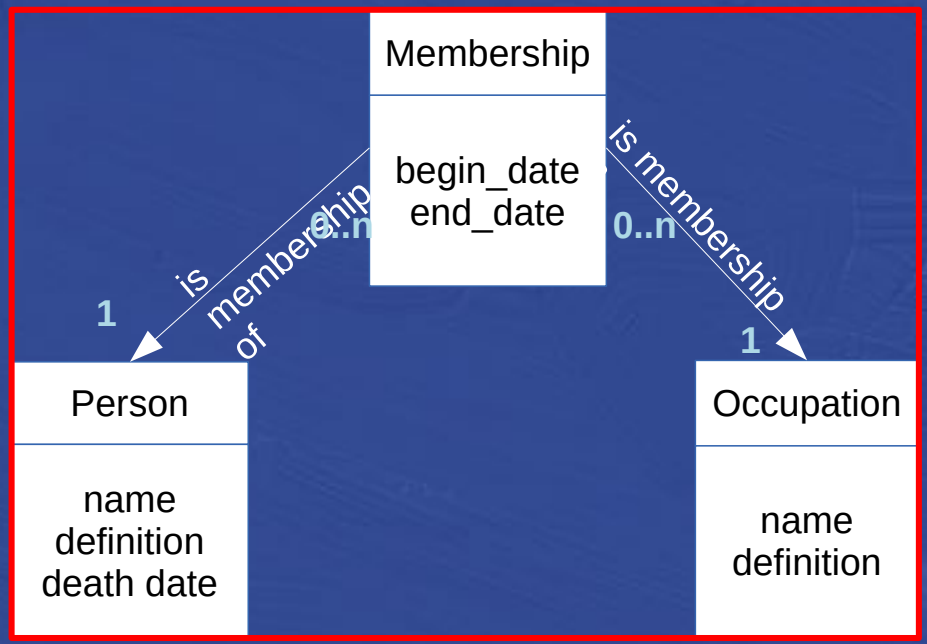
schema.org

?

Research agenda

Research specific data model

Research data



Foundational ontologies
& modelling best practices

DOLCE + Descriptions and Situations
& object-oriented modelling principles

Research agenda

Research specific data model

Research data

OntoClean

N. Guarino/C. A. Welty, « An Overview of OntoClean », in Steffen Staab, ed., Handbook on ontologies, 2nd ed. Berlin: Springer, 2009.

Essence (as defined by rigid intensional properties)
– A ‘student’ : not a class but a time-indexed property of a person

Identity (criteria to clearly distinguish individuals)
– The identity of a building as individual does not depend on its changing use

Unity (parthood)
– A crowd of humans vs a group having a plan

Dependence (one instance implies the existence of another one)
– No human without a birth and no birth without a human

Modelling Best Practices

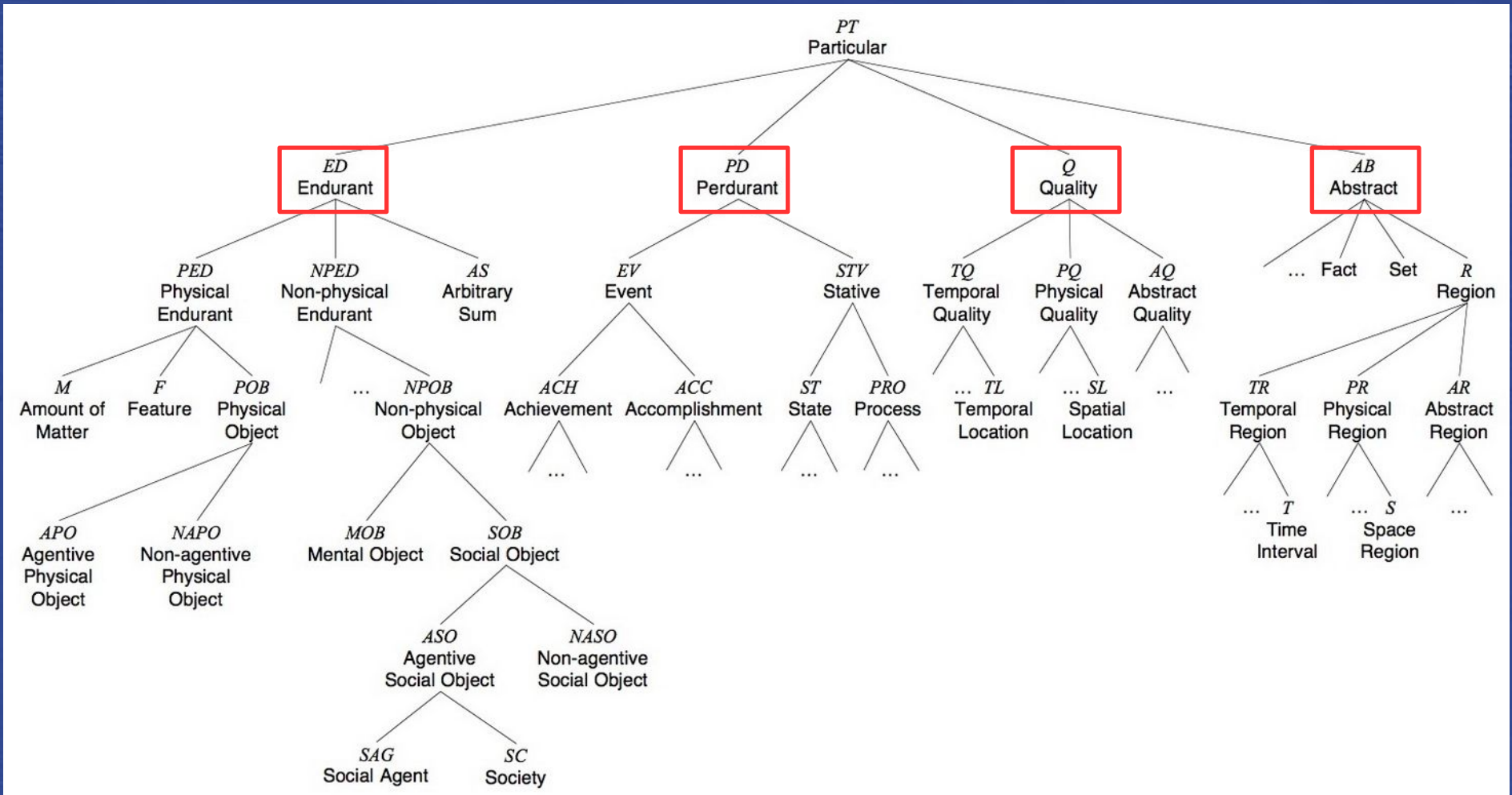
– Property inheritance, quantifiers, multiple statements on same properties
– Rich controlled vocabularies (researchers) but concise ontology (semantic engineers)

« **Foundational ontologies** are not directly used for applications; rather, they provide **conceptual handles** to solve cases of misunderstandings due to the limitations of expressiveness of the application languages. […]

DOLCE has remained fixed over the years fulfilling the purpose of top-level ontologies to provide a solid and stable basis for **modeling different domains**, in this way **ensuring interoperability** of reference and domain ontologies that use DOLCE. »

Borgo Stefano et al., « DOLCE: A descriptive ontology for linguistic and cognitive engineering1 », Applied Ontology, 18.11.2021, pp. 1-25.

Foundational ontologies
were developed to support the verification and improvement
of the **conceptualization** of a **domain of discourse.**



Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) – a foundational ontology designed in 2002 in the context of the WonderWeb EU project, developed by Nicola Guarino and his associates at the Laboratory for Applied Ontology (LOA) – WonderWeb Deliverable D18, p.14

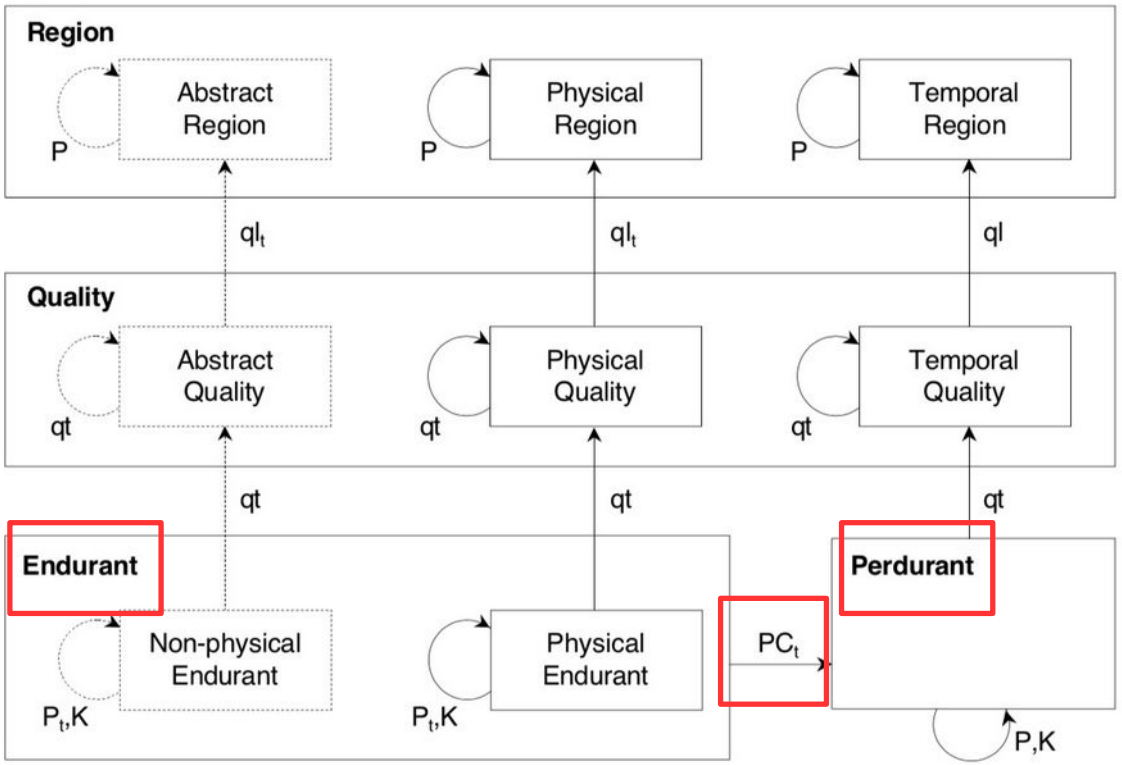


Figure 4: Primitive relations between basic categories (the dotted lines to the left indicate that we are less confident with what concerns non-physical endurants.

4.2.7 Participation

(Dd63) $\mathsf{PC}_{\mathsf{C}}(x,y) \triangleq \exists t(\mathsf{PRE}(y,t)) \wedge \forall t(\mathsf{PRE}(y,t) \rightarrow \mathsf{PC}(x,y,t))$ *(Const. Participation)*

(Dd64) $\mathsf{PC}_{\mathsf{T}}(x,y,t) \triangleq PD(y) \wedge \forall z((\mathsf{P}(z,y) \wedge \mathsf{PRE}(z,t)) \rightarrow \mathsf{PC}(x,z,t))$ *(Temporary Total Participation)*

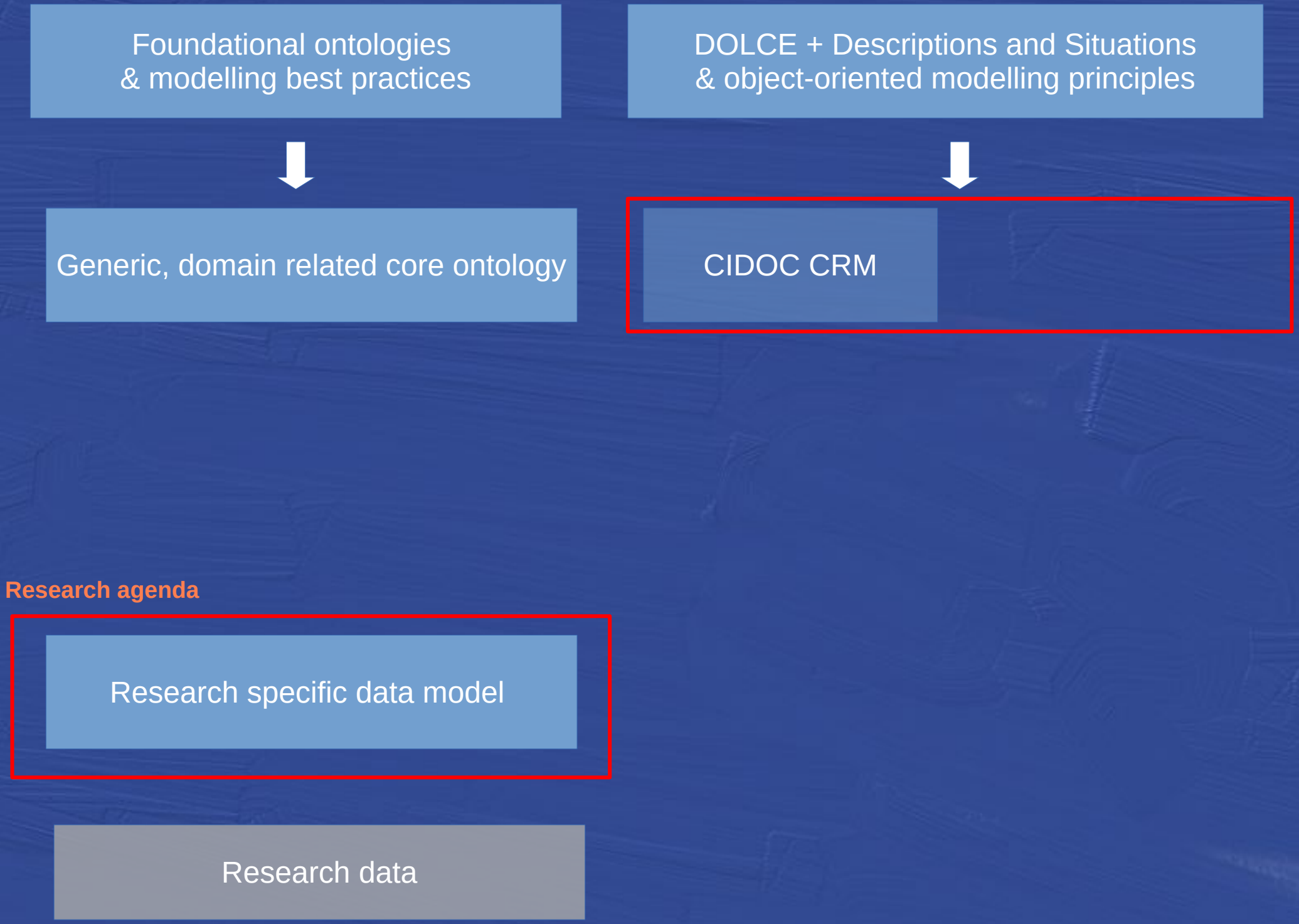
(Dd65) $\mathsf{PC}_{\mathsf{T}}(x,y) \triangleq \exists t(\mathsf{ql}_T(t,y) \wedge \mathsf{PC}_{\mathsf{T}}(x,y,t))$ *(Total Participation)*

(Dd66) $\mathsf{mpc}(x,y) \triangleq x = \sigma_t z(\mathsf{PC}_{\mathsf{T}}(z,y))$ *(Maximal Participant)*

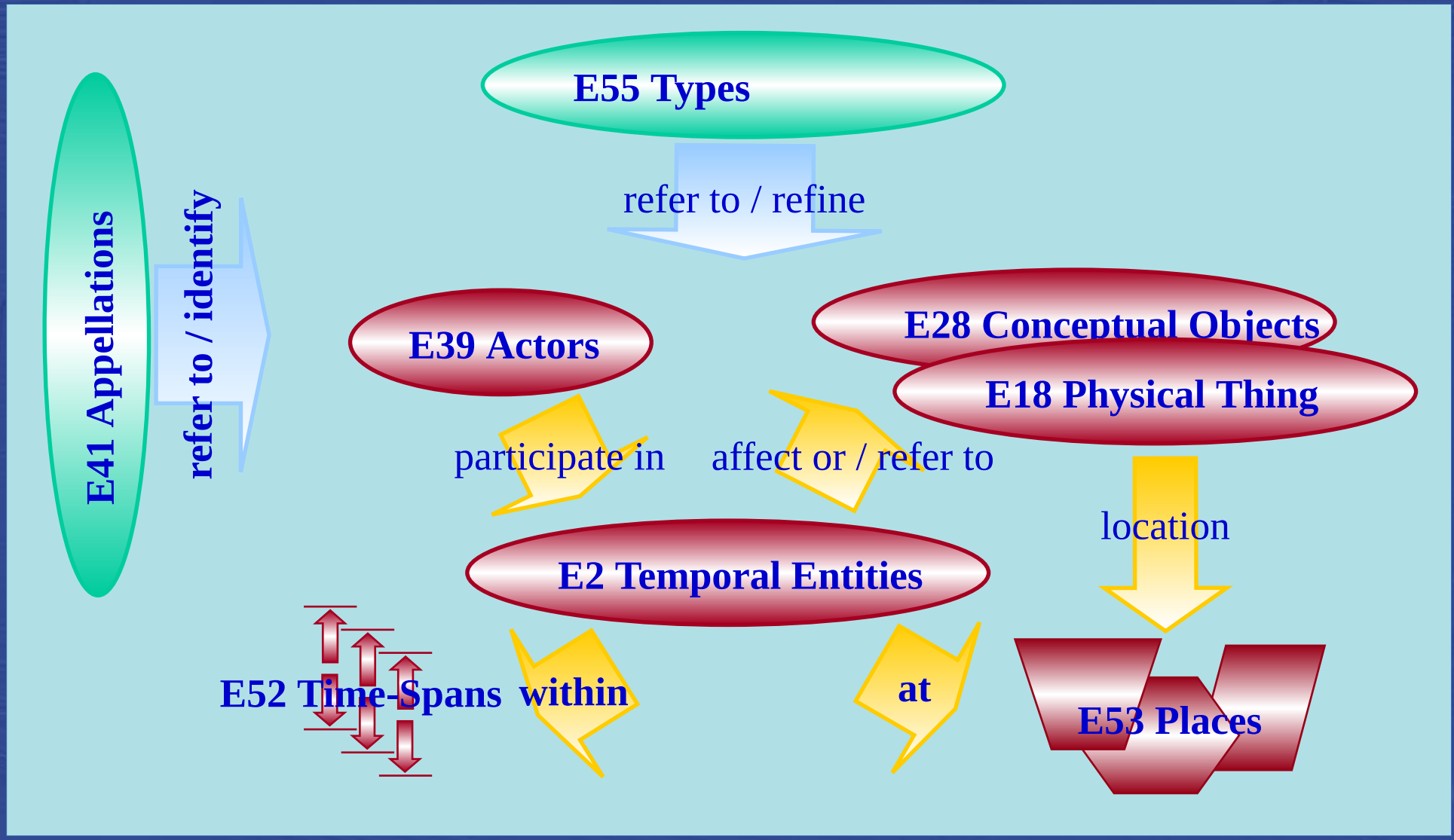
(Dd67) $\mathsf{mppc}(x,y) \triangleq x = \sigma_t z(\mathsf{PC}_{\mathsf{T}}(z,y) \wedge PED(z))$ *(Maximal Physical Participant)*

(Dd68) $\mathsf{lf}(x,y) \triangleq x = \sigma z(\mathsf{PC}_{\mathsf{T}}(y,z))$ *(Life)*

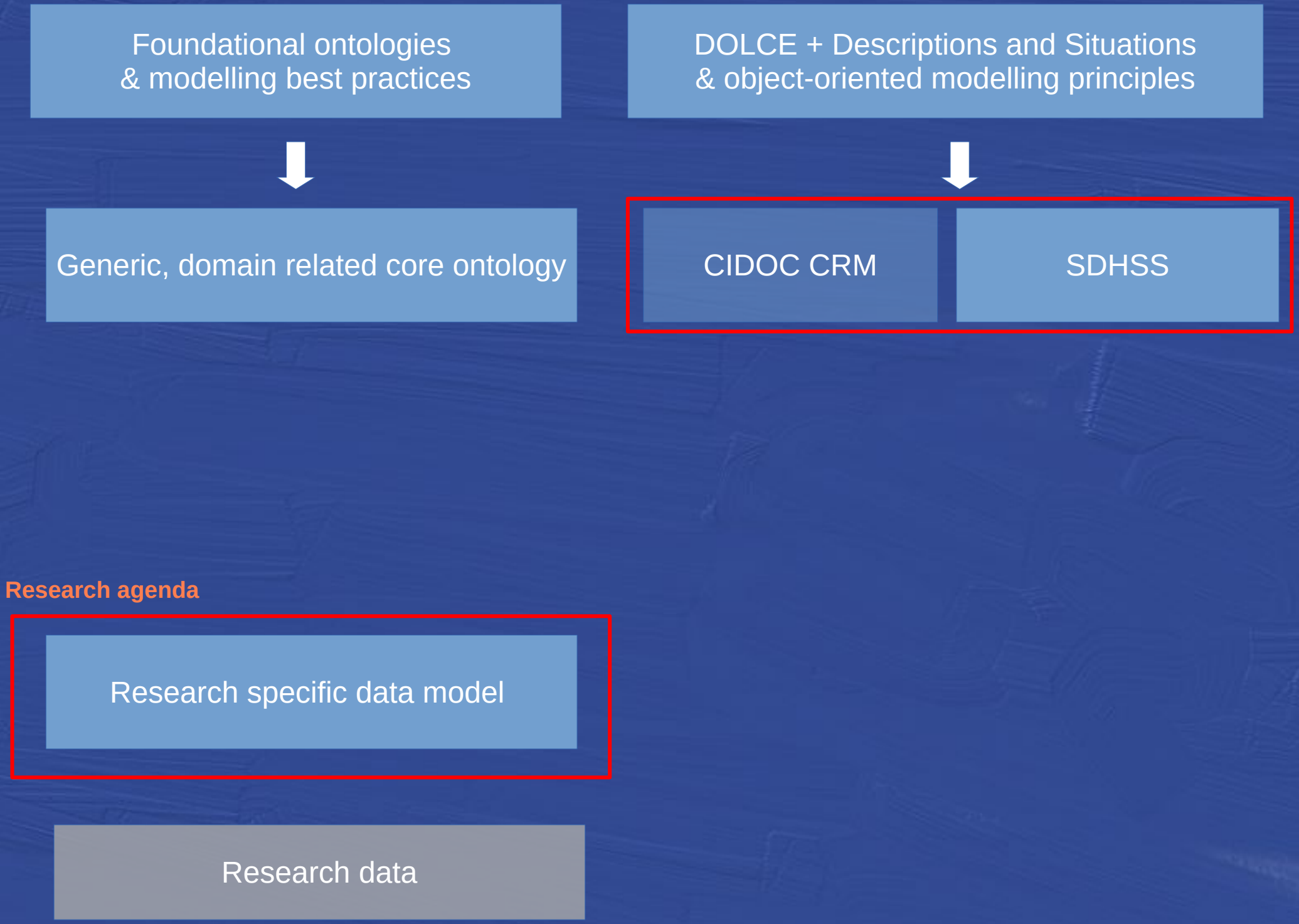
WonderWeb Deliverable D18, p.30

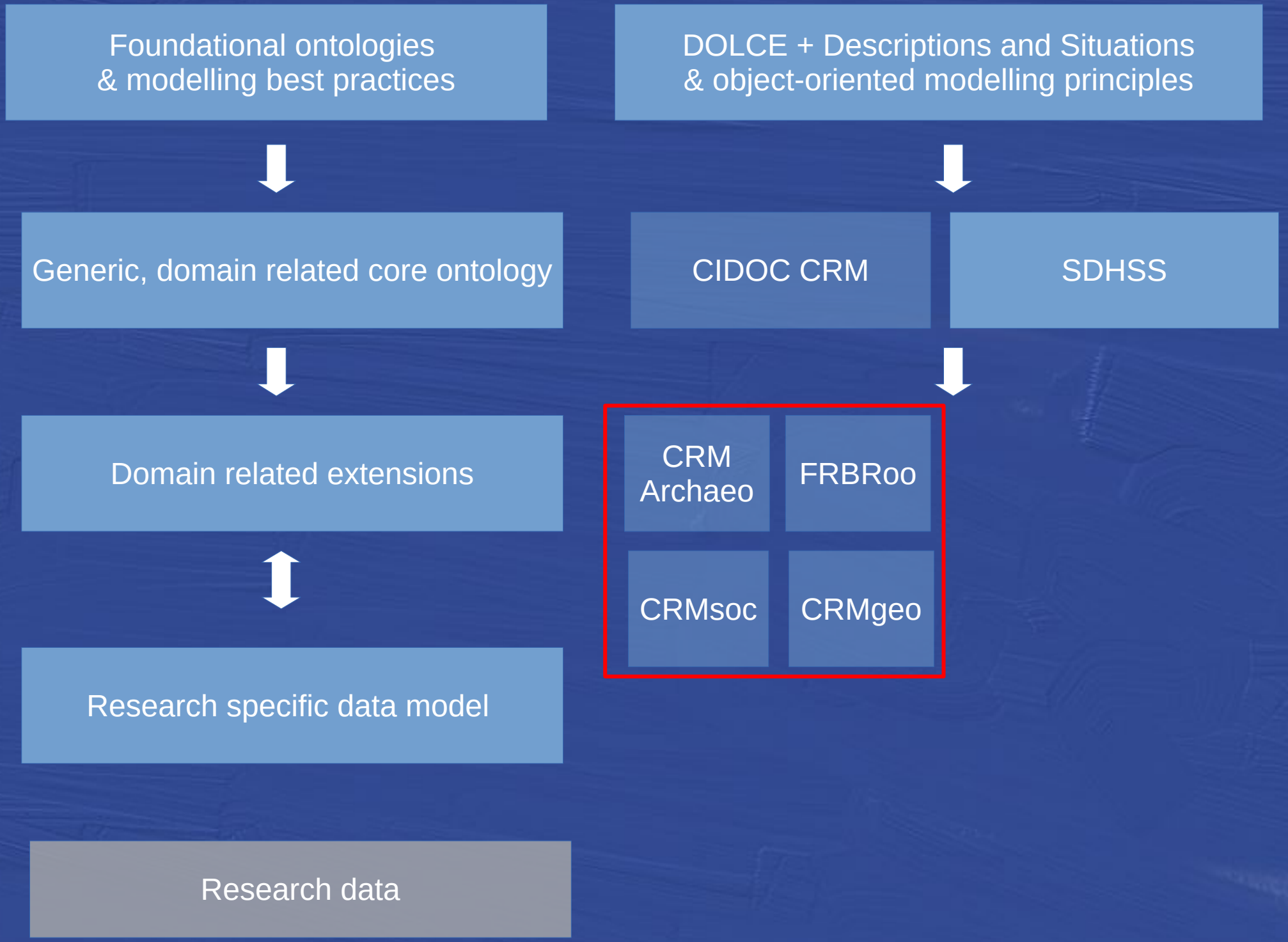

Foundational ontologies
& modelling best practices
DOLCE + Descriptions and Situations
& object-oriented modelling principles
Generic, domain related core ontology
CIDOC CRM
Research agenda
Research specific data model
Research data

The CIDOC CRM (ISO21127:2023 – community edition 7.1.3)
A semantic framework that provides *interoperability* between different sources of **cultural heritage information**



Stephen Stead (2008)


Foundational ontologies
& modelling best practices
DOLCE + Descriptions and Situations
& object-oriented modelling principles
Generic, domain related core ontology
CIDOC CRM
SDHSS
Research agenda
Research specific data model
Research data


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& modelling best practices
DOLCE + Descriptions and Situations
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Archaeo
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Research data

Semantic Data for Humanities and Social Sciences (SDHSS) CIDOC CRM Core Extension

Semantic Data for Humanities and Social Sciences (SDHSS) CIDOC CRM Top-Level Extension

Description:

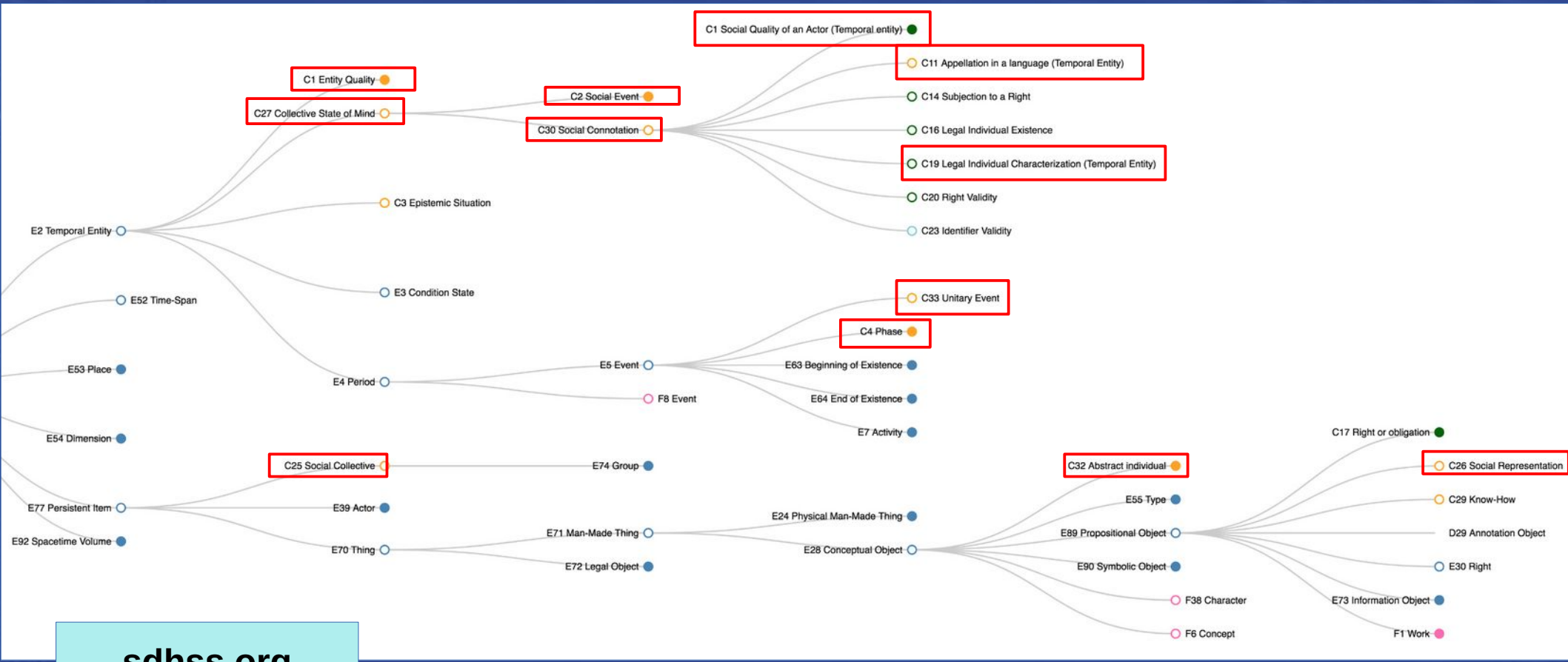
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The extension of CIDOC CRM for semantic data for humanities and social sciences (SDHSS) stems from the need to conceptualise the reality in the world, and more specifically factual information, from the point of view of historical research. The ontological commitment is therefore related to the domain of discourse of history but insofar as history, as a discipline that studies the life of humans and societies in the past, is interested in all the different aspects of social, economic, political, religious, literary and cultural life, the scope of this extension could be defined as the whole of social and human life, apprehended from the descriptive point of view, and global approach to reality, that characterises historical research.

This definition of the scope or domain modelled is based on the conviction that in a constructivist approach of scientific knowledge, a conceptualisation and data model can only be developed from the point of view of a specific discipline because *scientific objects* do not exist in the absolute but depend on the method and research agenda. They depend on the perspective or epistemic context researchers adopt in considering states of affairs: *scientific objects*, and semantic models modelling them, are not declared to be the only appropriate and exclusive representation of *things* in the pre-Kantian sense but defined as *intentional objects* constructed from the point of view of a discipline and methodological approach in relation to things in the world. Scientific objects are not the things in the world themselves, even if they must necessarily refer to them by way of observation or experimentation, if a scientific and therefore realistic approach is to be maintained. This corresponds to the notion of inter-objectivity in social sciences relying on the distinction between things in themselves and things as perceived, experienced and discussed by human subjects, in their shared intentionality and in relation to their social practices and context.

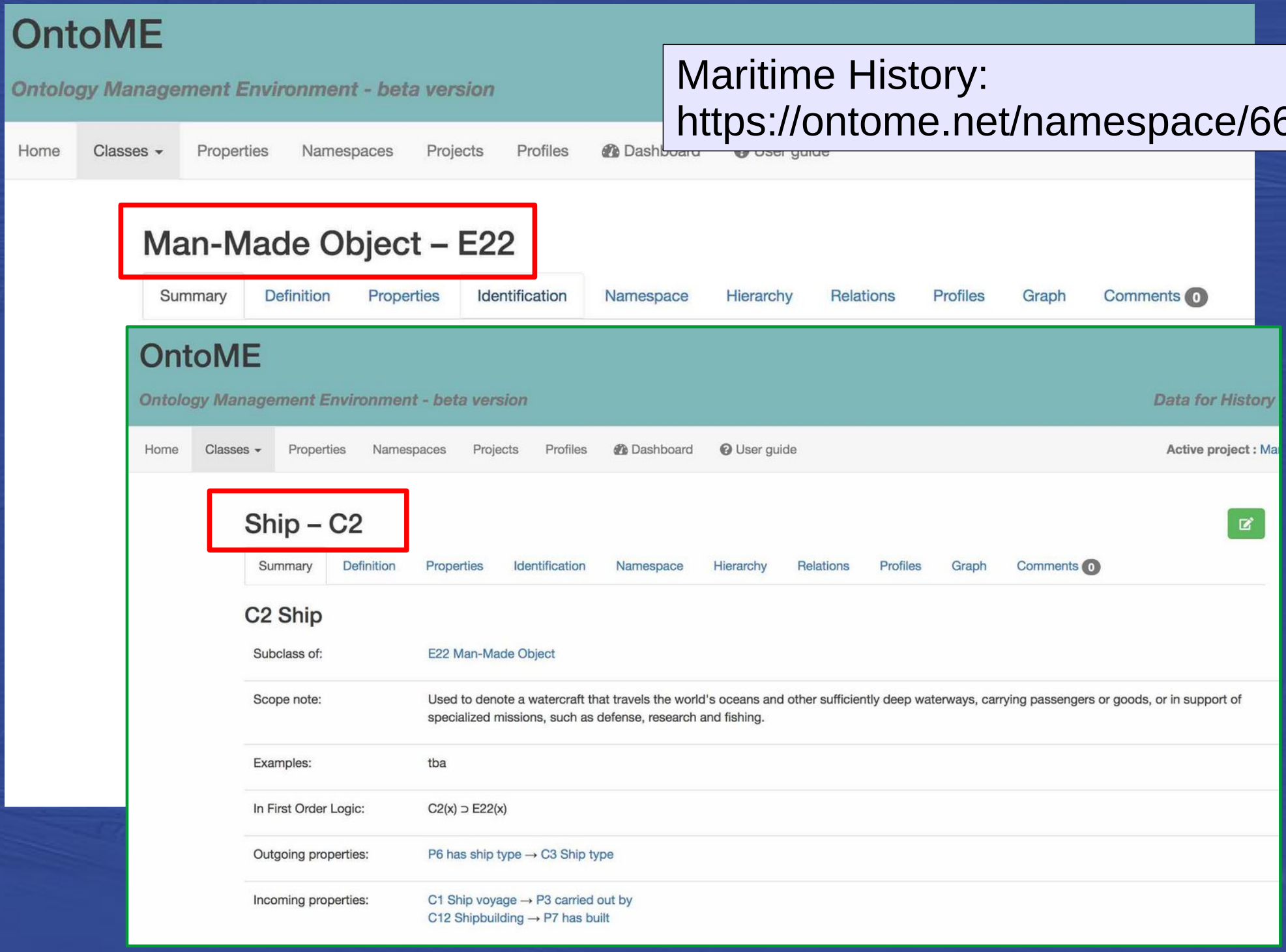
sdhss.org

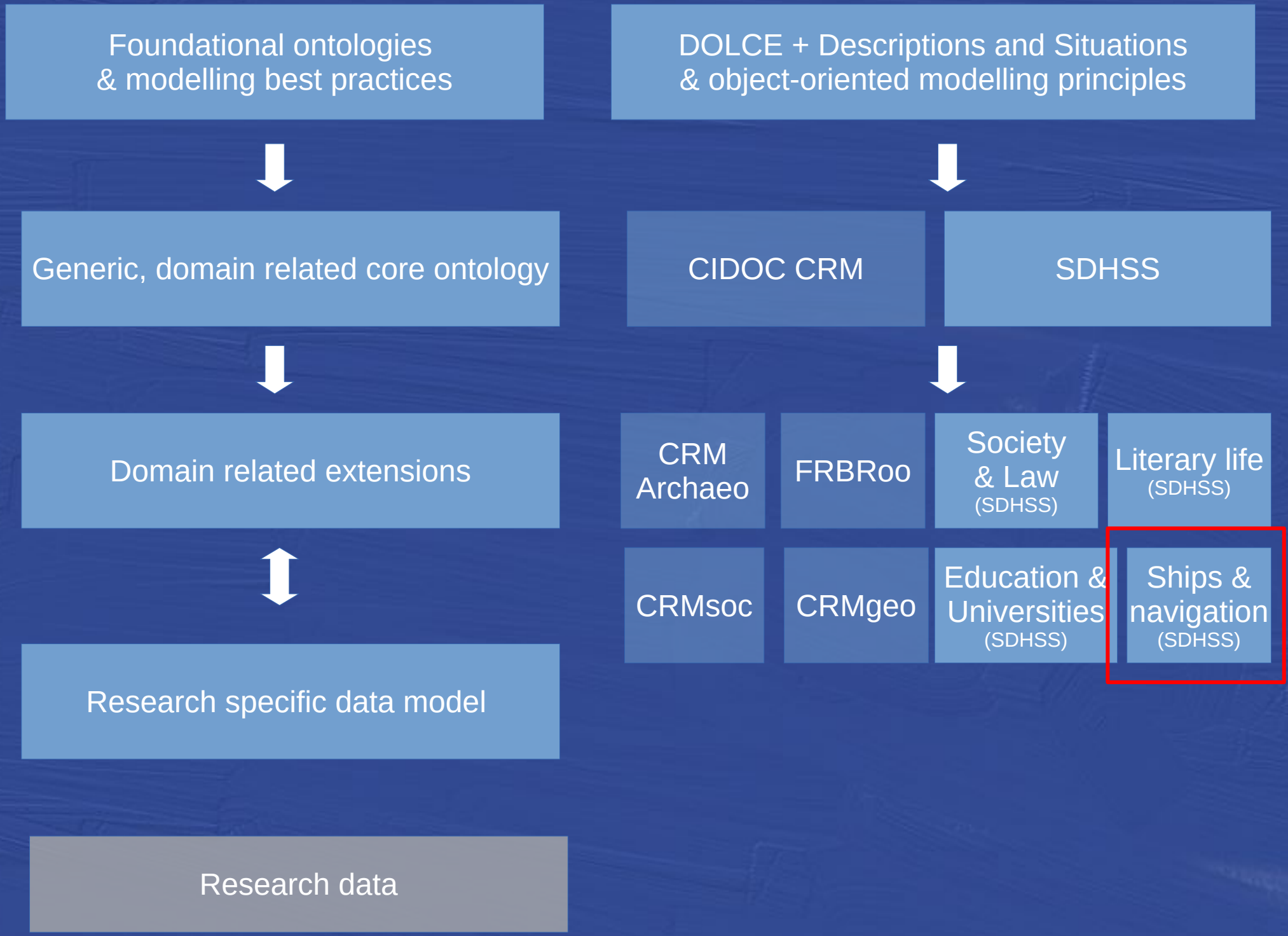
High level classes of a core extension of the CIDOC CRM that are suitable to express information related to HSS research agendas

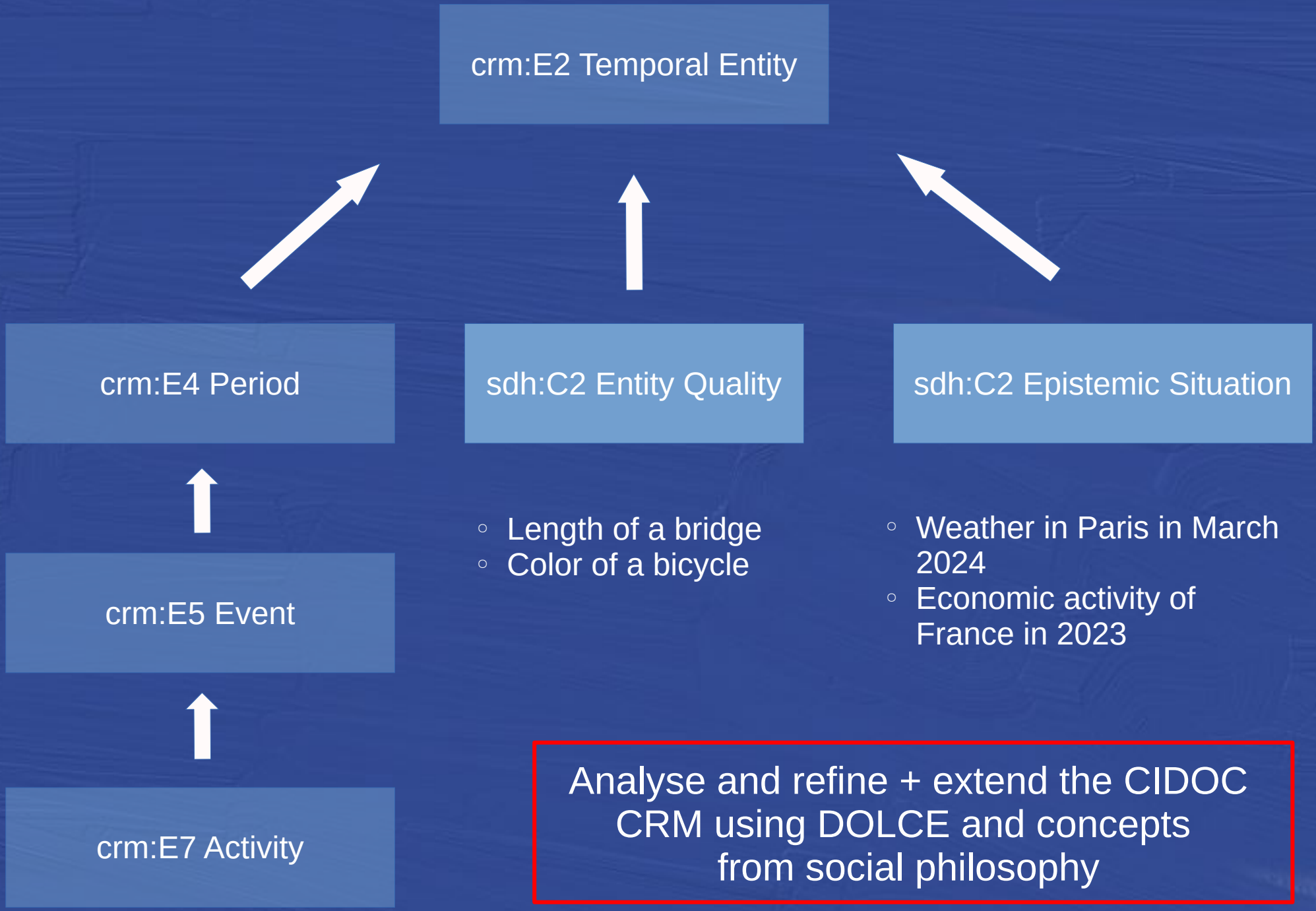


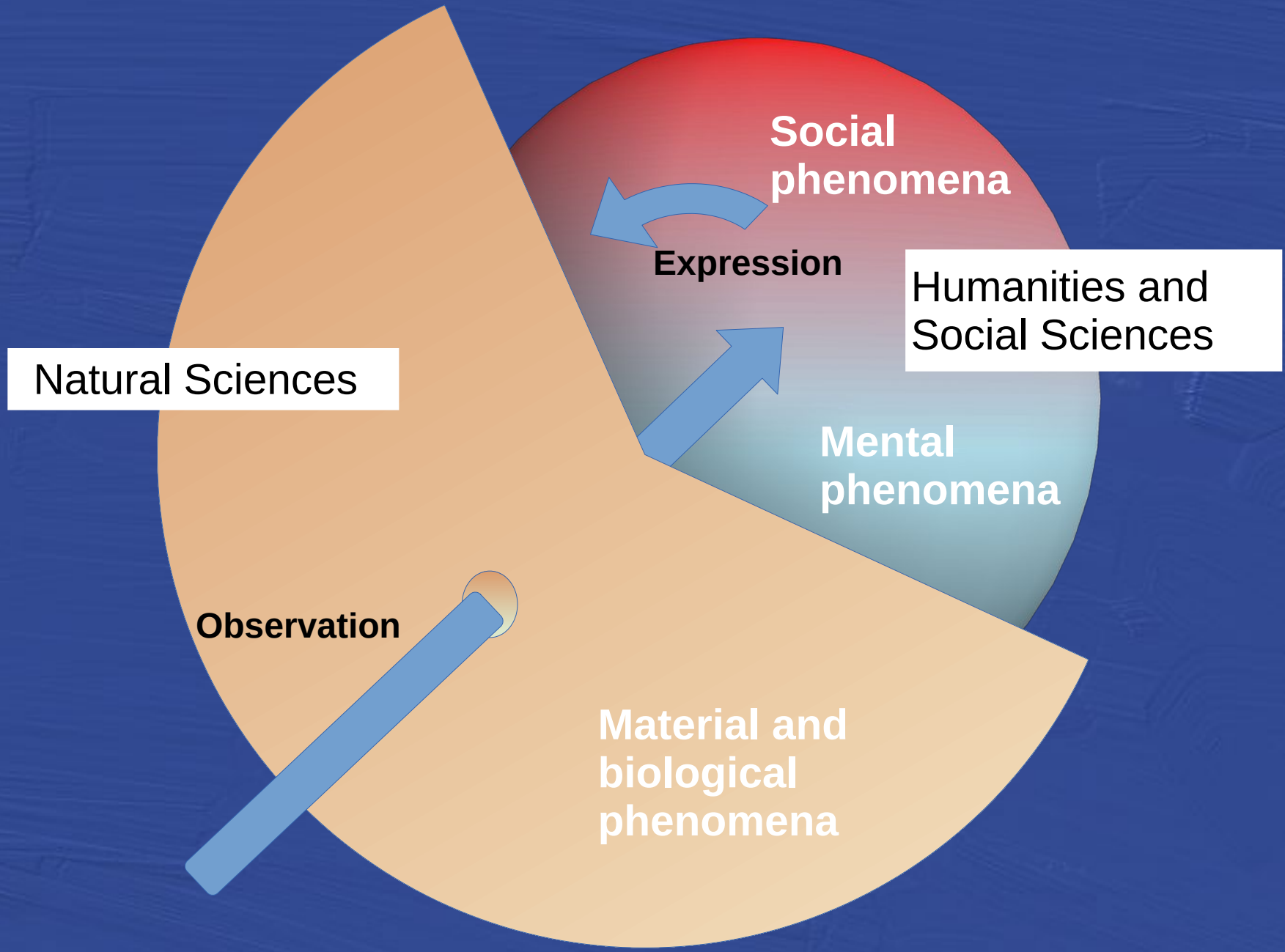
sdhss.org

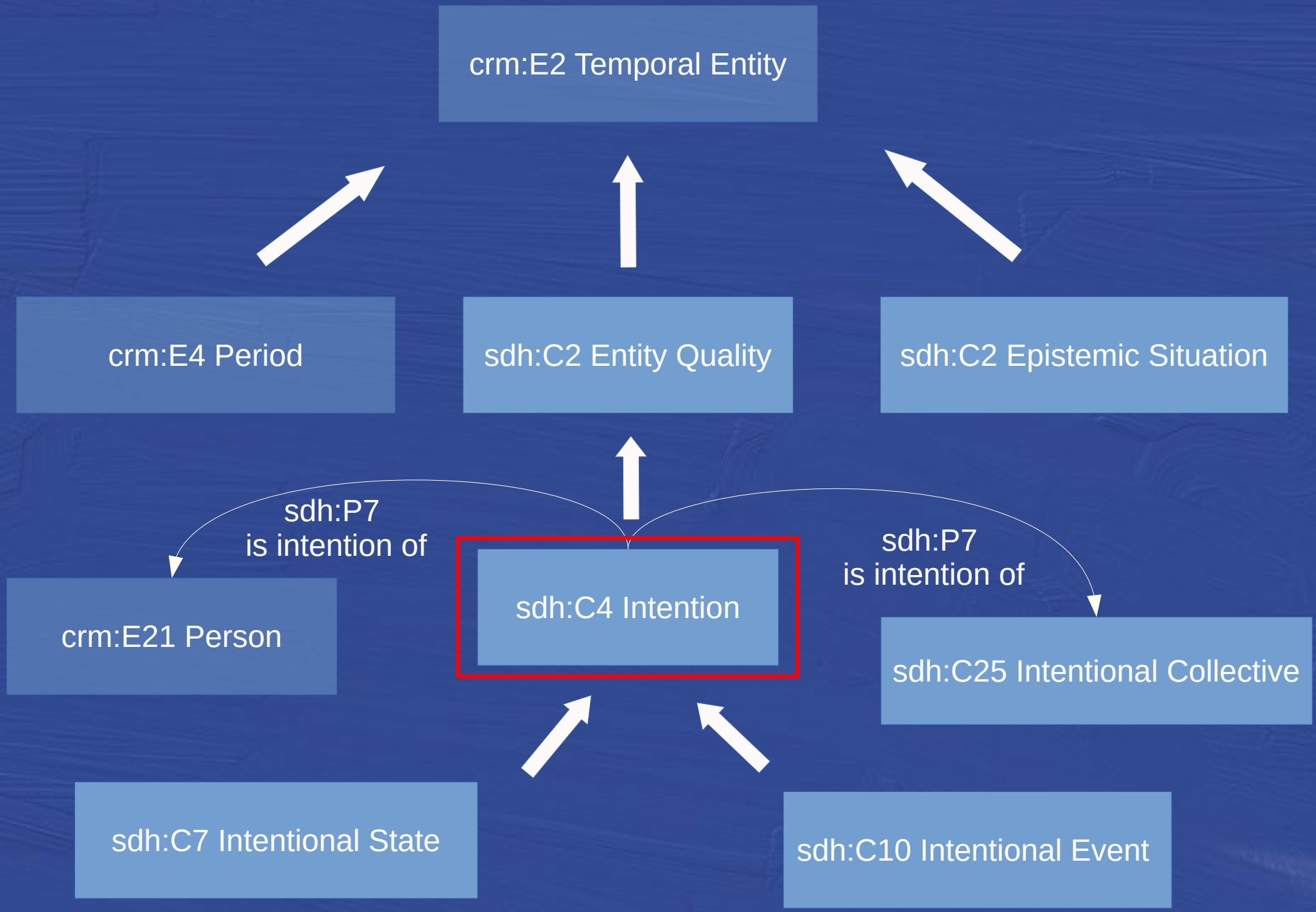
Maritime History:
https://ontome.net/namespace/66




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crm:E2 Temporal Entity
crm:E4 Period
sdh:C2 Entity Quality
sdh:C2 Epistemic Situation
crm:E5 Event
Length of a bridge
Color of a bicycle
Weather in Paris in March 2024
Economic activity of France in 2023
crm:E7 Activity
Analyse and refine + extend the CIDOC CRM using DOLCE and concepts from social philosophy


Social phenomena
Expression
Humanities and Social Sciences
Natural Sciences
Mental phenomena
Observation
Material and biological phenomena


crm:E2 Temporal Entity
crm:E4 Period
sdh:C2 Entity Quality
sdh:C2 Epistemic Situation
sdh:P7
is intention of
sdh:C4 Intention
sdh:P7
is intention of
crm:E21 Person
sdh:C25 Intentional Collective
sdh:C7 Intentional State
sdh:C10 Intentional Event





Social representations

Individual minds

Social representations

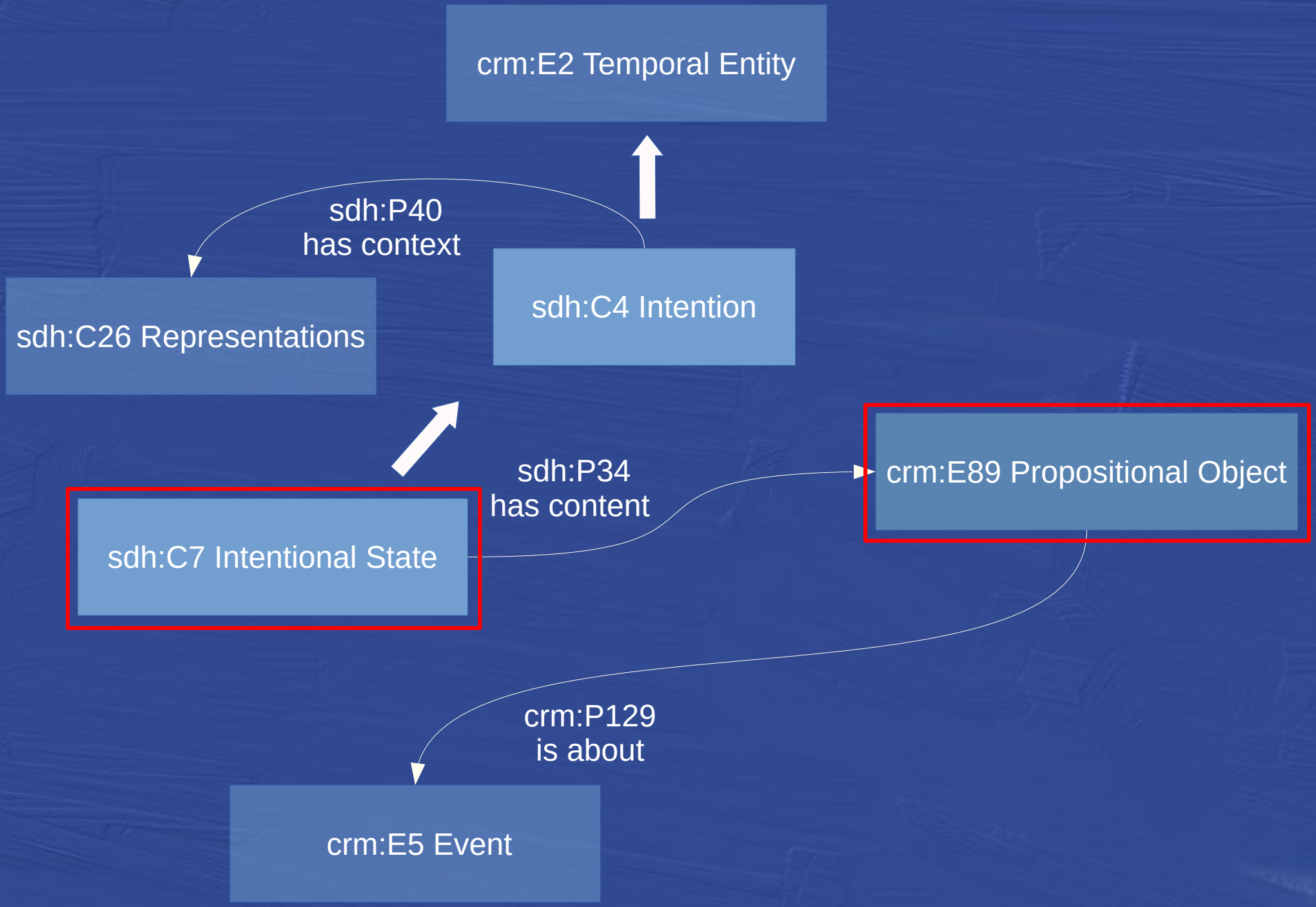
Individual minds



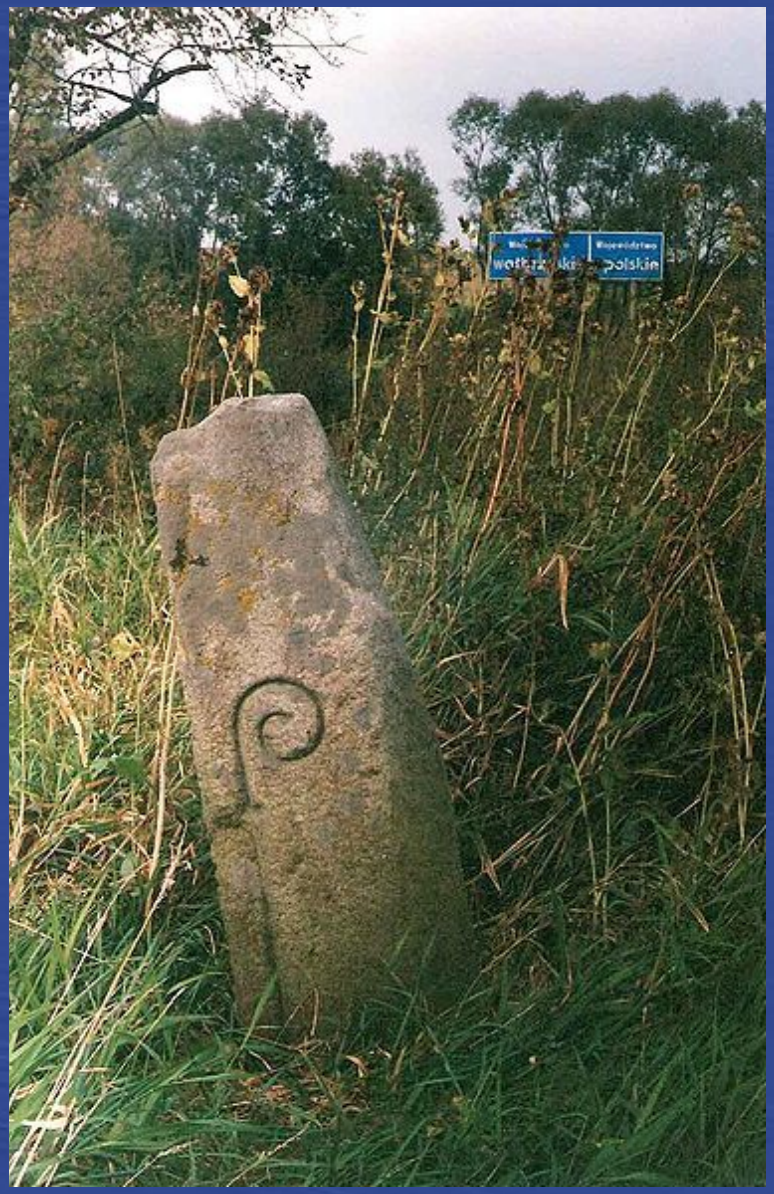
Different perspectives on the same events: interpreting a car race as fun or as a crime



Sports car photo created by azerbaijan_stockers - www.freepik.com


crm:E2 Temporal Entity
sdh:P40
has context
sdh:C26 Representations
sdh:C4 Intention
sdh:C7 Intentional State
sdh:P34
has content
crm:E89 Propositional Object
crm:P129
is about
crm:E5 Event

Artefacts and their social significance: a bondary stone that defines a boundary



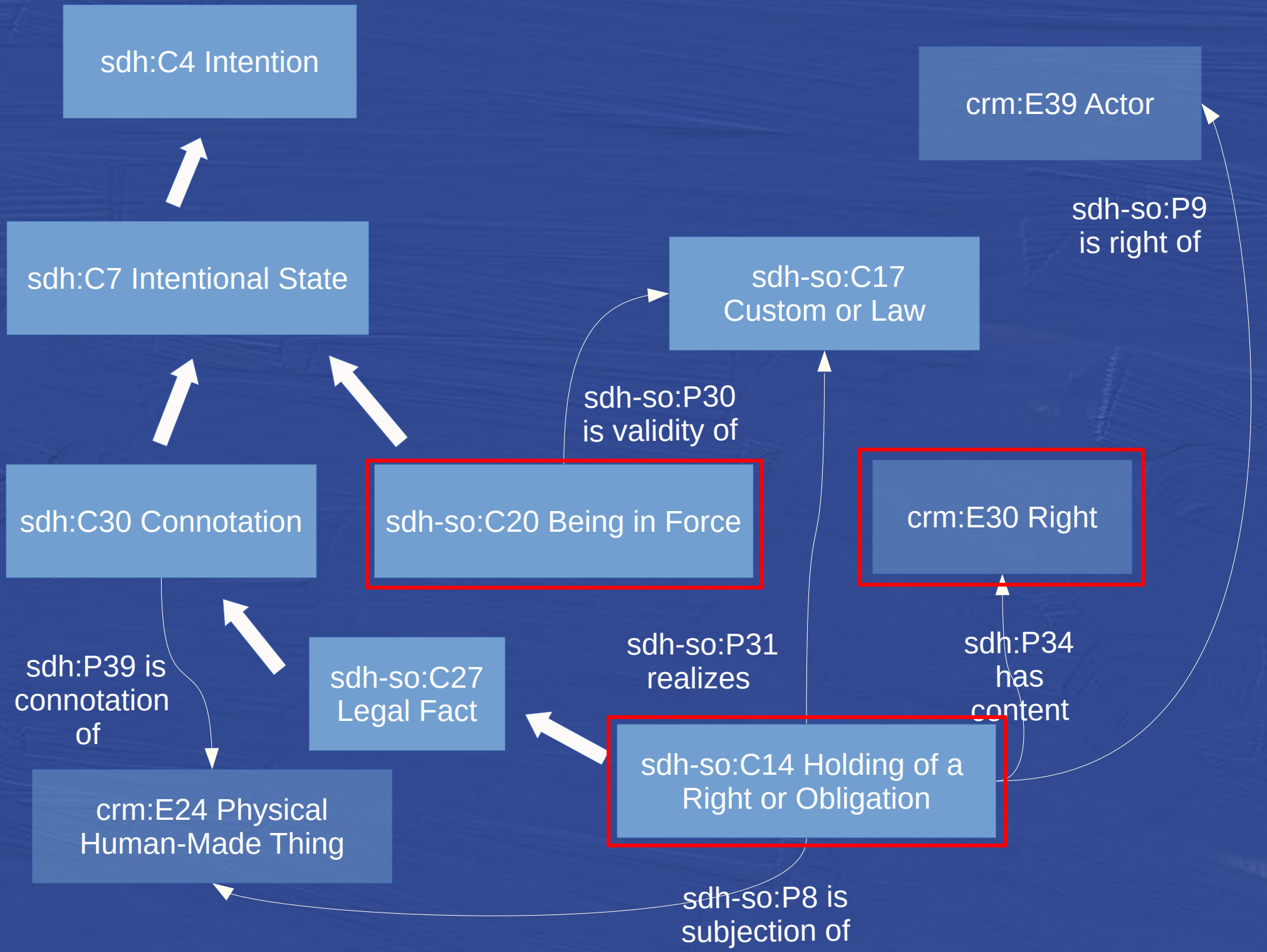
Border stone from c. 1280 on the border of the lands of the Wrocław bishops.

In the background the contemporary road sign marking the border between Opole and Wałbrzych voivodships.

Photo: Władysław Łoś (CC By SA 3.0)

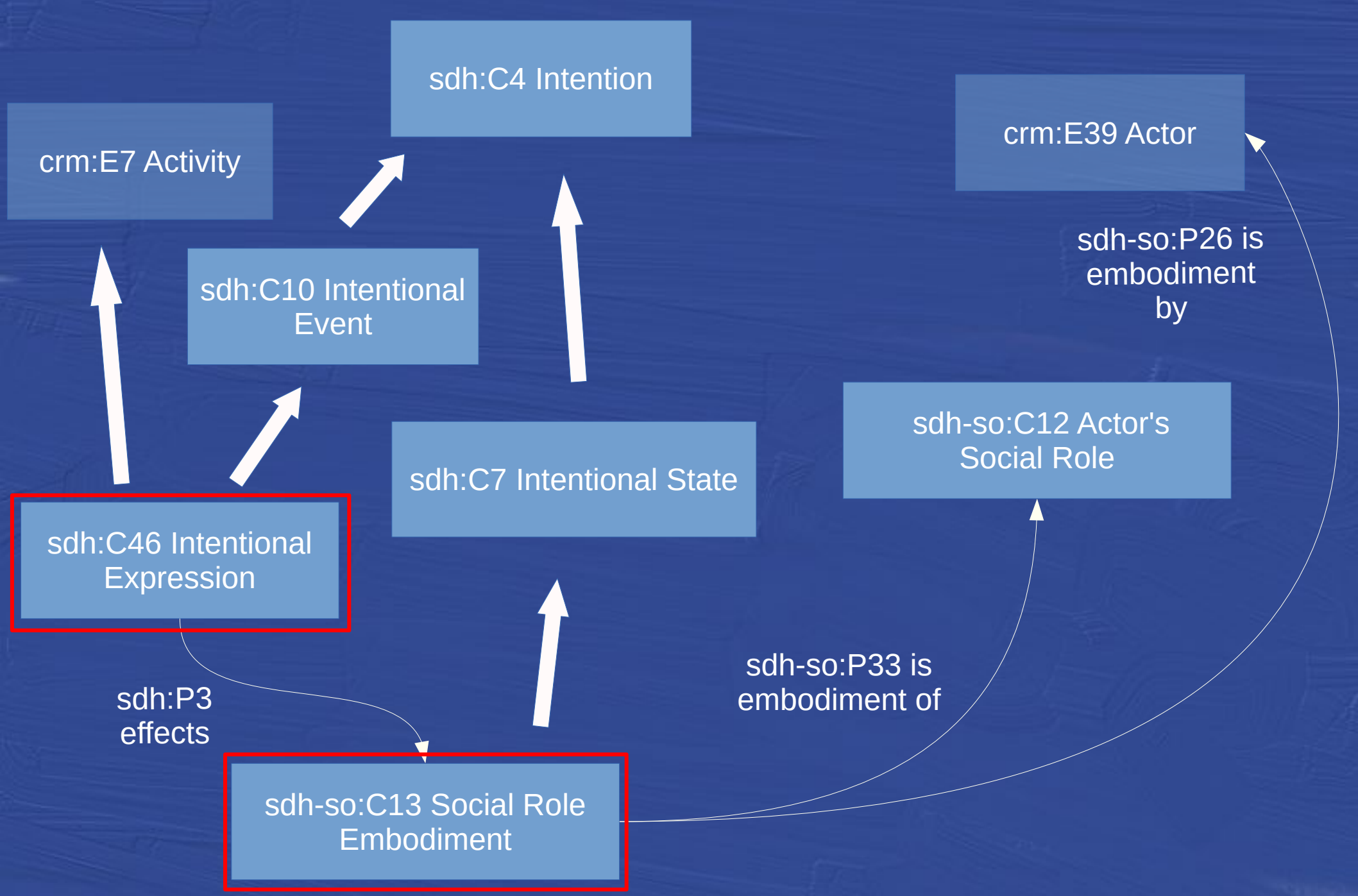
The physical object, the notion of border, the symbol of the episcopal crozier, the function of bishop.

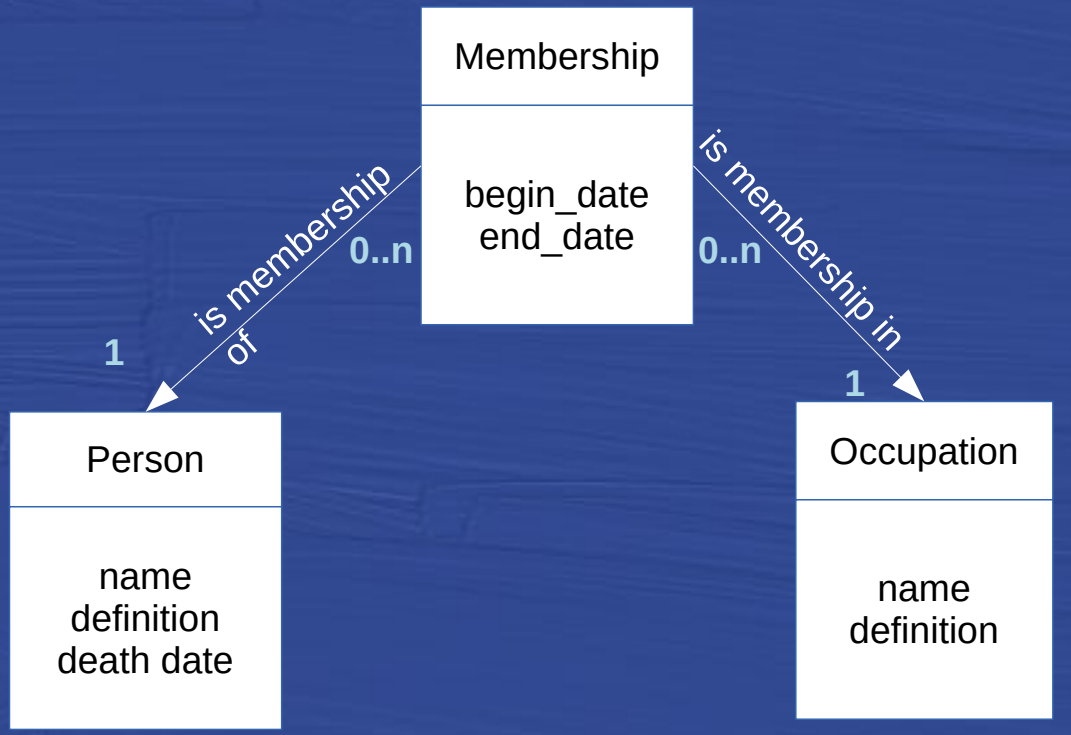
The bio-physical world, the social world


sdh:C4 Intention
crm:E39 Actor
sdh:C7 Intentional State
sdh-so:P9
is right of
sdh-so:C17
Custom or Law
sdh-so:P30
is validity of
sdh:C30 Connotation
sdh-so:C20 Being in Force
crm:E30 Right
sdh:P39 is
connotation
of
sdh-so:C27
Legal Fact
sdh-so:P31
realizes
sdh:P34
has
content
sdh-so:C14 Holding of a
Right or Obligation
crm:E24 Physical
Human-Made Thing
sdh-so:P8 is
subjection of

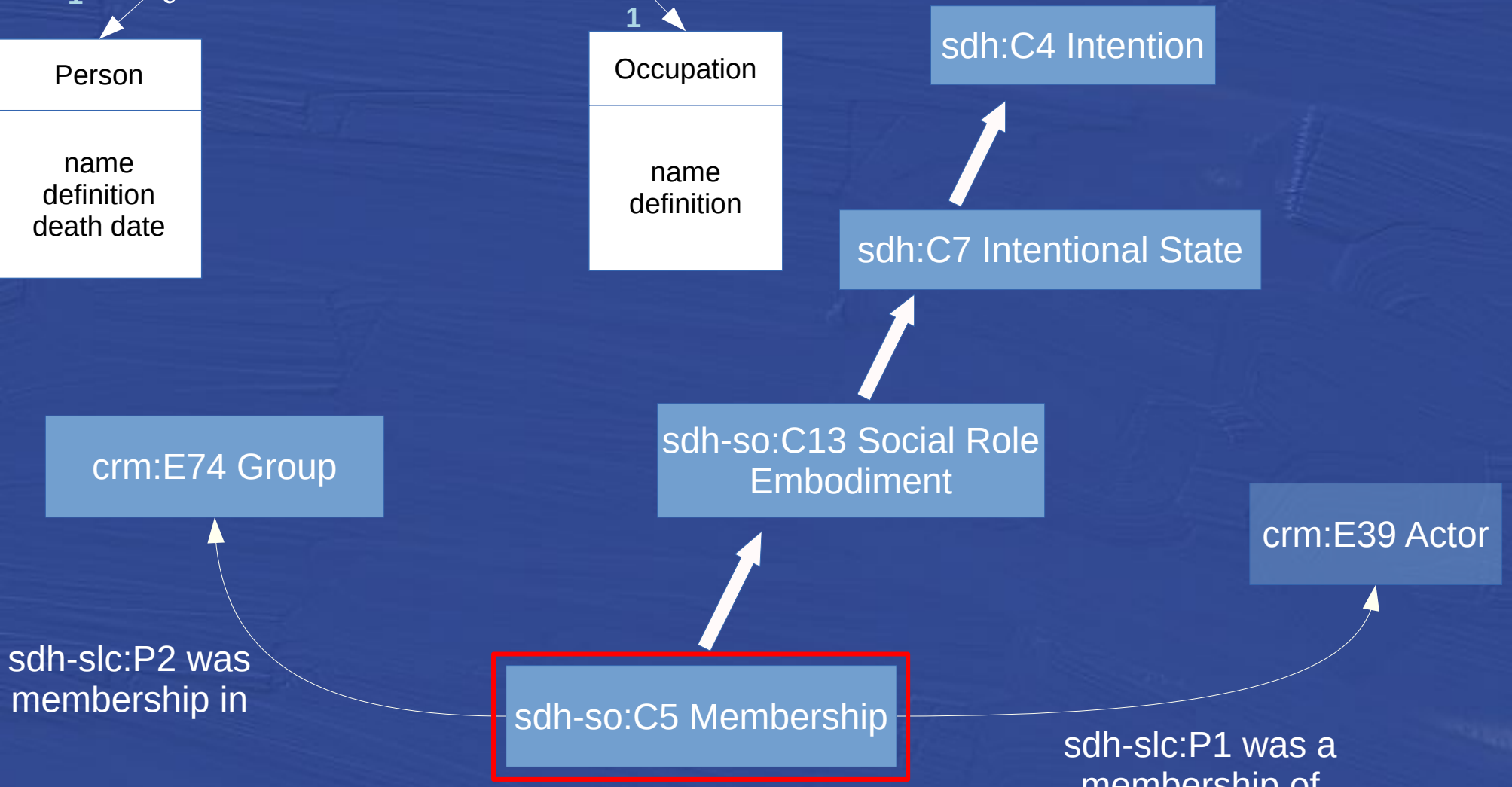
Social roles: the rite that establishes the function

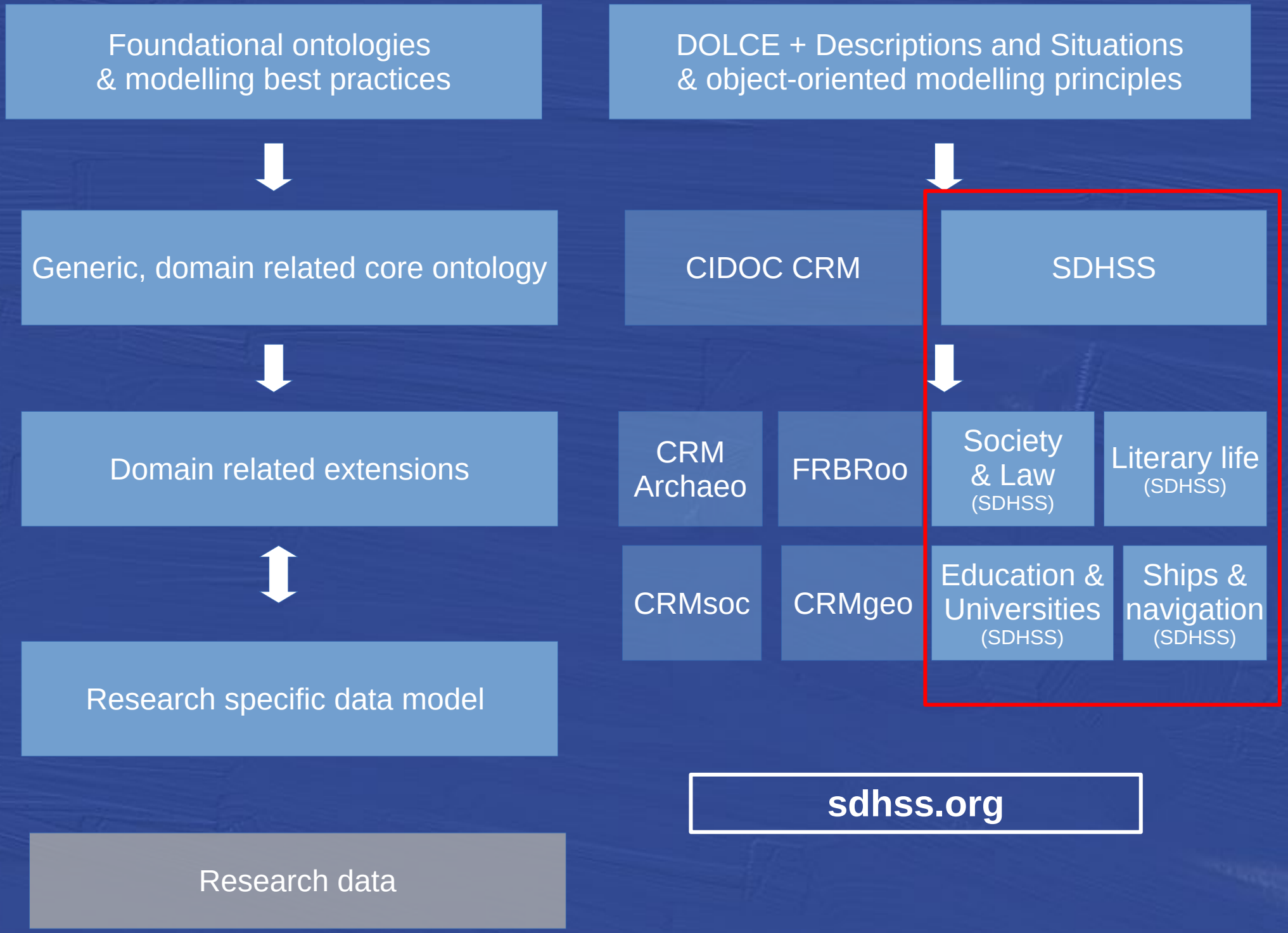


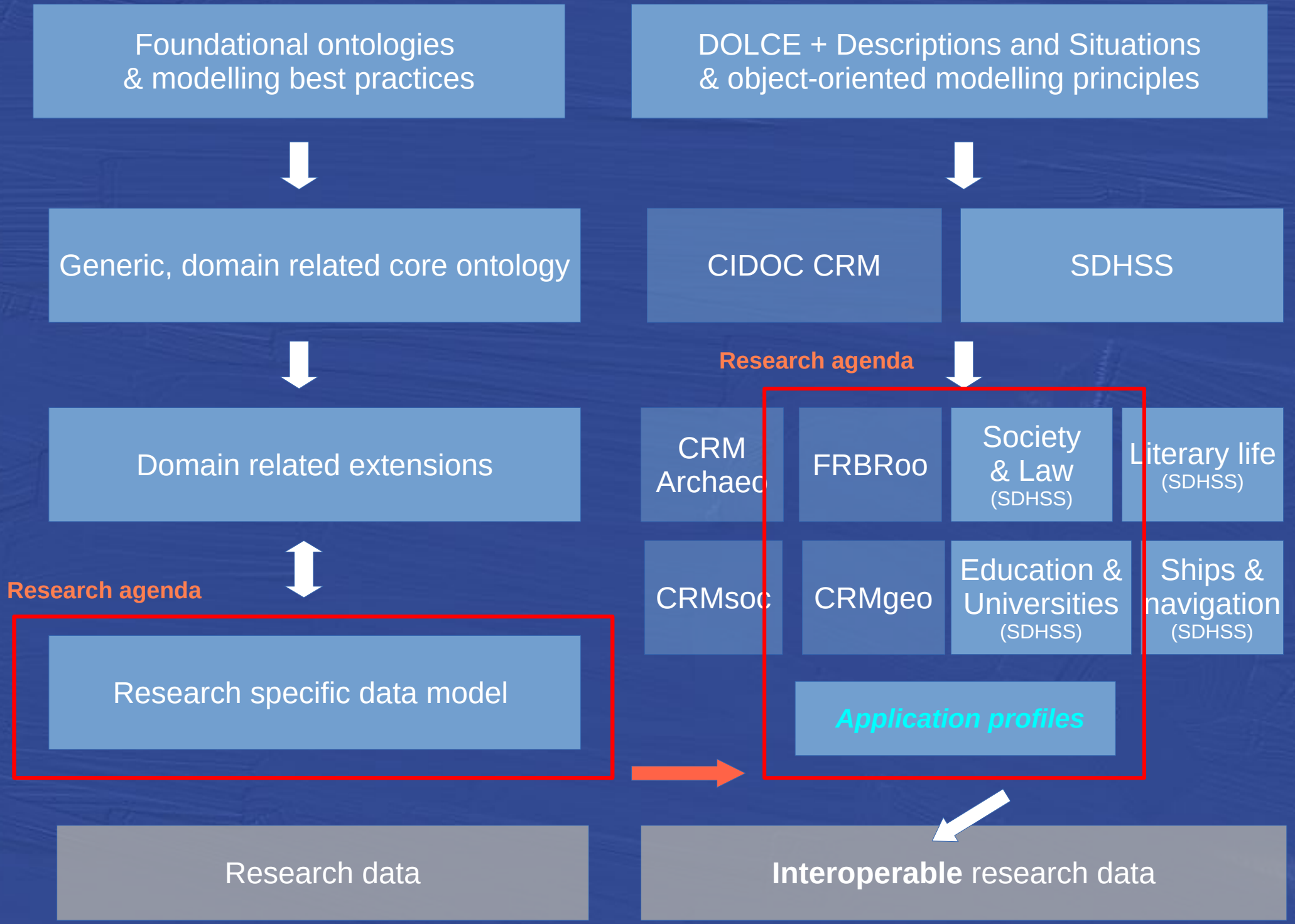

sdh:C4 Intention
crm:E7 Activity
crm:E39 Actor
sdh:C10 Intentional Event
sdh-so:P26 is embodiment by
sdh-so:C12 Actor's Social Role
sdh:C7 Intentional State
sdh:C46 Intentional Expression
sdh:P3 effects
sdh-so:P33 is embodiment of
sdh-so:C13 Social Role Embodiment



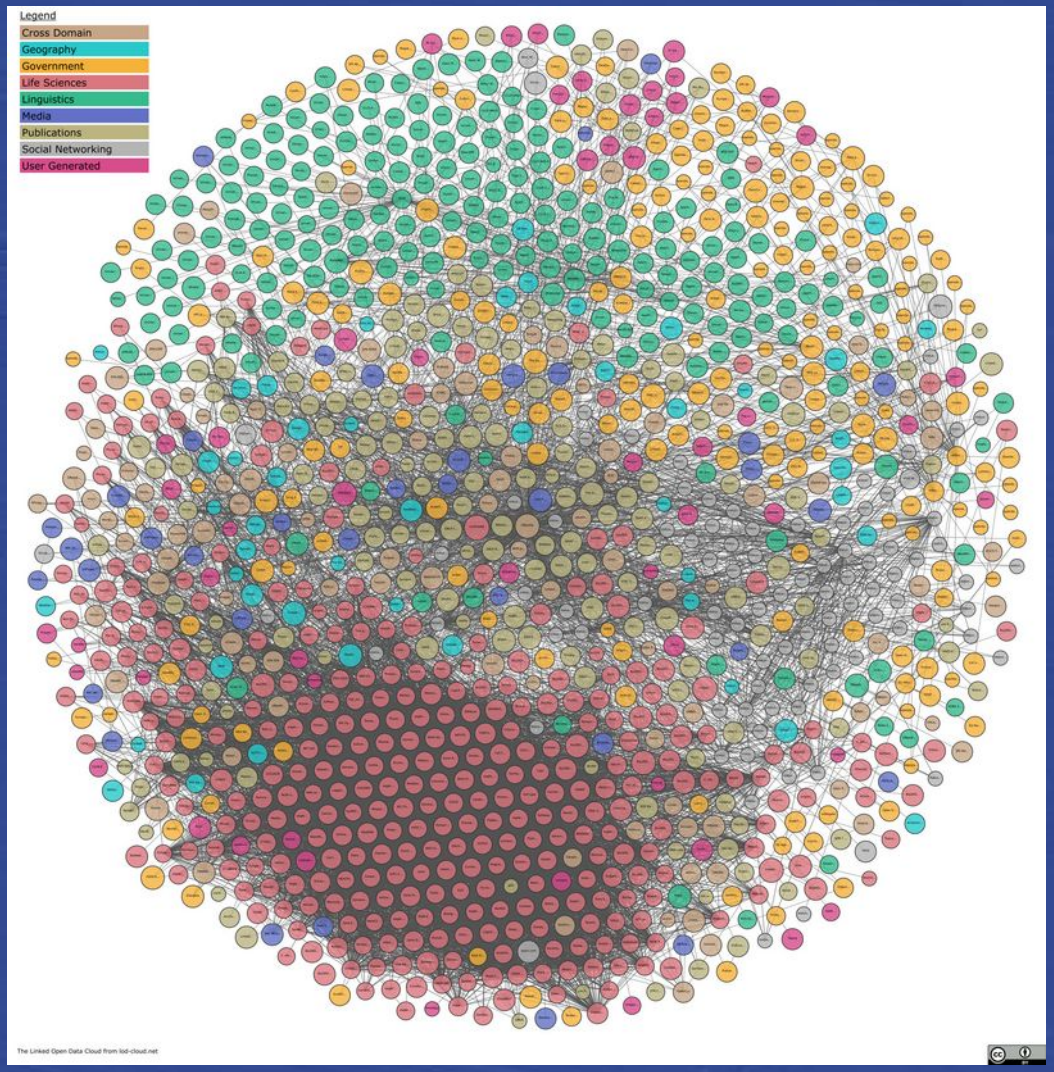
"An ontology is
a *formal explicit specification*
of a shared conceptualization
of a domain of interest"




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Application profiles
Research data
Interoperable research data

A giant knowledge graph of FAIR ORD easily and directly reusable by HSS



https://lod-cloud.net/